PROBABILISTIC LISTENER: A CASE OF REFLEXIVE *ZIJI* “SELF” AMBIGUITY RESOLUTION IN MANDARIN CHINESE

An Honors Thesis Presented

By

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ABSTRACT

How to account for variation in the resolution of ambiguous pronouns across languages and context is a major contemporary challenge in psycholinguistics. Probabilistic models are promising attempts to understand the underlying mechanism to process ambiguous pronouns in many previous literature. This study further explores the ability of two probabilistic models --- Bayesian models and Rational Speech Act (RSA) models --- to explain cross-linguistic variation in pronoun interpretation preferences by looking at a different language --- Mandarin Chinese. The binding property of the Chinese Mandarin *ziji* "self" represents the challenge of pronoun interpretation. An example is shown below, where the reference of the reflexive *ziji* “self” is ambiguous, permitting both a local and a long-distance construal represented by the fact that *ziji* is coindexed with both *Zhang Wei* (with the subscript i) and *Xiao Ming* (with the subscript j).[Zhang Wei]$_i$ says [Xiao Ming]$_j$ BA [ziji]$_{i/j}$ confuse$_{PERF}$.`[Zhang Wei]$_i$ says that [Xiao Ming]$_j$ confused [self]$_{i/j}$.' Meanwhile, this study also investigates the interpretation variation across context by attempting to apply the models to account for the different interpretation in different context. Specifically, I look at item-level variation when building Bayesian and RSA models instead of consistently using a uniform parameters in the models for all items as is typically done in experimental studies on pronoun interpretation. In short, the present study tests the Bayesian models and RSA models in a different language (Mandarin Chinese) with a novel set of pronouns (reflexive pronouns), predicting item-level interpretation variation. New experimental data show a very good fit of the model predictions for *ziji* pronoun interpretation preferences. However, some experimental data on the interpretation of another ambiguous pronoun *ta* which is an alternative pronoun to *ziji* deviate from model predictions in certain conditions.

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Author Note

I am very grateful to the following people, without whom I would not have been able to complete this research and without whom I would not have been able to complete my undergraduate degrees at UMass Amherst.

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make my voice heard and my actions seen in the near future, like a firefly lighting itself up in the dark of night. If a better future has not yet arrived, I can be the only brightness for myself and for others.

“愿中国青年都摆脱冷气，只是向上走，不必听自暴自弃者流的话。能做事的做事，能发声的发声。有一分热，发一分光，就令萤火一般，也可以在黑暗里发一点光，不必等候炬火。此后如竟没有炬火：我便是唯一的光。”

Abstract

How to account for variation in the resolution of ambiguous pronouns across languages and context is a major contemporary challenge in psycholinguistics. Probabilistic models are promising attempts to understand the underlying mechanism to process ambiguous pronouns in many previous literature. This study further explores the ability of two probabilistic models — Bayesian models and Rational Speech Act (RSA) models — to explain cross-linguistic variation in pronoun interpretation preferences by looking at a different language — Mandarin Chinese. The binding property of the Chinese Mandarin *ziji* "self" represents the challenge of pronoun interpretation. An example is shown below, where the reference of the reflexive ziji "self" is ambiguous, permitting both a local and a long-distance construal represented by the fact that *ziji* is coindexed with both *Zhang Wei* (with the subscript i) and *Xiao Ming* (with the subscript j).

(1) [张 伟]$_i$ 说 [小 明]$_j$ 把 [自己]$_{i/j}$ 弄糊涂了。
[Zhang Wei]$_i$ says [Xiao Ming]$_j$ BA [ziji]$_{i/j}$ confuse$_{PERF}$.
'[Zhang Wei]$_i$ says that [Xiao Ming]$_j$ confused [self]$_{i/j}$.'

Meanwhile, this study also investigates the interpretation variation across context by attempting to apply the models to account for the different interpretation in different context. Specifically, I look at item-level variation when building Bayesian and RSA models instead of consistently using a uniform parameters in the models for all items as is typically done in experimental studies on pronoun interpretation. In short, the present study tests the Bayesian models and RSA models in a different language (Mandarin Chinese) with a novel set of pronouns (reflexive pronouns), predicting item-level interpretation variation. New experimental data show a very good fit of the model predictions for *ziji* pronoun interpretation preferences. However, some experimental data on the interpretation of another ambiguous pronoun *ta* which is an alternative pronoun to *ziji* deviate from model predictions in certain conditions.

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1 Introduction

Language as a human system of communication is instructions to communicate ideas, opinions, thoughts, etc. Speech, gestures, and expressions are the embodiment of language in the human body. Although all natural languages have a set of encoding and decoding rules, languages are able to integrate these finite and discrete rules to generate infinite utterances and meanings to achieve different social goals. Meanwhile, interpreting language in the context of communication is not just like simply decoding combinatorial structure: one utterance is possible to be interpreted in multiple ways depending on factors such as the communicators' world knowledge, the relation between communicators, the context of the communication, etc. Studying how humans interpret utterances might shed some light on general principles of human cognition. This research picks the interpretation of pronouns as a special window to examine ways to account for humans' underlying processing mechanisms.

One puzzle in pronoun interpretation is how and why interpretation varies across languages. One example from (Schulz, Burnett, & Hemforth, 2021) that represents this puzzle is (2). The ambiguous pronoun can refer back either to the subject (the postman) or the object (the streetsweeper) of the matrix clause. If you ask native speakers of English or German for the referent of the pronoun, most will tell you that the referent is the subject (the postman). However, if you ask native speakers of French, most will tell you the referent is the object of the clause (the streetsweeper) (Schulz et al., 2021). A series of studies has shown evidence indication a general preference for the first-mentioned antecedent and for the subject (Järvikivi, van Gompel, Hyönä, & Bertram, 2005; Hemforth et al., 2010) in the resolution of ambiguous non-reflexive pronouns in sentence like (2), but these accounts are not universal and cannot explain French speakers' preference for the object of the clause.

(2) The postman$_i$ met the streetsweeper$_j$ before **he**$_{i/j}$ went home.

Another puzzle in pronoun interpretation is how and why interpretation varies across contexts. One example adopted from (Kehler & Rohde, 2013) that represents this puzzle is shown in (3). Pronouns in all three sentences can refer to either Mitt or Rick. Native English speakers have the shared intuition that the pronoun in (3-a) refers to Mitt, the pronoun in (3-b) refers to Rick, and the pronoun in (3-c) refers to Rick. The preferences in (3-a) and (3-b) can be explained by a theory that listeners like to assign a pronoun to the person mentioned in the subject position of the first clause. However, this cannot explain the interpretation preference in (3-c). Instead, it could possibly be explained by the claim that pronoun interpretation is determined by world knowledge since the loser is more likely to demand a recount based on world experience.

(3) a. Mitt narrowly defeated Rick, and the press promptly followed **him** to the next primary state.

b. Rick was narrowly defeated by Mitt, and the press promptly followed **him** to the next primary state.

c. Mitt narrowly defeated Rick, and **he** quickly demanded a recount.

The inconsistency and complexity in pronoun interpretation preferences above raise several interesting questions. Where do these intuitions come from? How can people quickly and reliably decide the referent among all alternatives? What rules or mechanisms does the brain use to do so? Several accounts/heuristics were mentioned above such as the general preference for the first-mentioned antecedent and for the subject, and the world knowledge bias. However, there is no single heuristic that can account for all these interpretation data (Kehler & Rohde, 2013). Another framework theory for pronoun interpretation is presented by Grice which highlights the role of conversational implicatures in language understanding. The core of Grice's proposal was a set of conversational maxims (Grice, 1975). The maxim of quantity says that the speaker tries to use the pronoun that gives as much information as it needed, and no more. The maxim of quality

says that the speaker follows grammar rules and does not give any false information. The maxim of manner says that the speaker avoids ambiguity. Gricean listeners rely on assuming that speakers try their best to exploit these maxims to choose the best utterances to convey particular meanings, thus work backward to disambiguate speakers' utterances. However, this set of theoretical principles also encountered difficulties in the experimental psycholinguistic literature (Goodman & Frank, 2016). Meanwhile, this kind of informal theory cannot make directional, qualitative predictions and is not capable to capture the quantitative nature of experimental data.

Probabilistic models are another attempt to understand the underlying mechanism to process ambiguous pronouns. The way these models work is by applying Bayes' rule, calculating the probability or conditional probability of each uncertain event to describe listeners' inferences (Kehler & Rohde, 2013). Sources of uncertainty include the listener's world knowledge, the speaker's choice of utterances among all alternatives, and the listener's choice of interpretation among all alternatives. I will call the basic probabilistic models **Bayesian Models** that only uses Bayes' rule in the rest of this manuscript. Building on the Bayes' rule in the basic Bayesian Models, **Rational Speech Act (RSA) Models** are another kind of more complicated probabilistic models (Frank & Goodman, 2012). The RSA framework views communication as recursive reasoning between a rational speaker and a rational listener, and formalizes this view using the tools of game theory, information theory, and Bayesian inference. Recursive reasoning means that in the interpretation process, listeners assume the speakers have chosen their utterances to maximize the utility (the utility of an utterance is how well it communicates a certain meaning to a listener) they want to gain, so that listeners can update their pronoun interpretation via Bayes' Rule. There have been many experimental studies to test these probabilistic models and showed that these models are promising in resolving the puzzles presented in previous paragraphs. (Kehler & Rohde, 2013; Frank & Goodman, 2012; Schulz et al., 2021). More details on these two models and on previous literature will be

introduced later in the subsections.

1.1 Bayesian Models

Kehler and Rohde (2013) suggests that the listener's interpretation of a pronoun in a communication is given by Bayesian inference: the listener's possibility of interpreting a pronoun as a specific referent is proportional to their prior world beliefs that how likely this referent will be mentioned next and the possibility of the speaker's possibility of producing this pronoun to signal a specific referent. This relationship between pronoun production and interpretation is show in Equation (4).

(4) $$P_L(referent \mid utterance) \propto P(referent) \times P_S(utterance \mid referent)$$

There are three terms involved in this equation. The first term $P_L(referent \mid utterance)$ on the left-hand side of the equation is called **posterior**. It represents the possibility for the listener to decide on the referent, given that they heard a pronoun being spoken. The second term $P(referent)$ on the right-hand side of the equation is called **prior**. It represents the listeners' prior world knowledge on which referent is more likely to be mentioned in the specific communication context, regardless of the referring expression used. The last term $P_S(utterance \mid referent)$ on the right-hand side of the equation is called **likelihood**. It represents the possibility of the speaker uttering a certain pronoun, given that they knew which referent they want to refer to. Therefore, in the Bayesian model, a listener's interpretation bias relies jointly on the prior knowledge that a particular referent will be more likely to be mentioned than other alternatives under a specific context and on the likelihood that the speaker would have chosen a specific pronoun to convey the information of that referent. The Bayesian Model predicts that if independent estimates of the prior, likelihood, and posterior probabilities were obtained, Equation (4) would approximately hold.

The Bayesian Model could be a solution to the puzzles raised above because it is a

generalization of the relationship and is not limited to a certain language or a certain context (unlike the heuristic that there exists a general preference for the for the subject in English). In other words, the model sets up a big picture of pronoun interpretation and allows internal adjustments by changing corresponding parameters in the model to fit different languages or contexts.

Kehler and Rohde (2013) applies the Bayesian model to explain the interpretation preference for subject in pronoun interpretation process in the context of subject-biased implicit causality verbs like *amaze*, as well as to explain the different strength of the subject preference in passive and active voice. They conducted an experiment using subject-biased implicit causality contexts in a 2x2 manipulation, shown in (5). (5-a) and (5-b) have pronoun prompts, while (5-c) and (5-d) are in the no-pronoun prompt conditions. (5-a) and (5-c) are in active voice while (5-b) and (5-d) are in passive voice.

(5) a. Amanda amazed Brittany. She _____.

b. Brittany was amazed by Amanda. She _____.

c. Amanda amazed Brittany. _____.

d. Brittany was amazed by Amanda. _____.

They estimate the values for the terms on the right-hand side of Equation (4) from data collected in experiments in the no-pronoun prompt conditions ((5-c) and (5-d)) to predict pronoun interpretation for the left-hand side, modeled by $P_L(referent \mid utterance)$. If the Bayesian model is correct, the predicted pronoun interpretation should correlated with the actual interpretation that are estimated directly in the pronoun-prompt conditions of the experiment ((5-a) and (5-b)). A comparison between the Bayesian model prediction and actual interpretation biases is shown in Table 1. The table shows that the Bayesian model not only predicts the subject preferences in both active and passive voice successfully, but also captures the fact that the active voice has a stronger subject preference than the passive voice. The predicted and actual values are highly correlated,

confirming the predictions of the Bayesian model.

Table 1
Bayesian Model Prediction and Actual Biases for subject antecedent example

	Predicted	Actual
Active	81%	74%
Passive	59%	60%

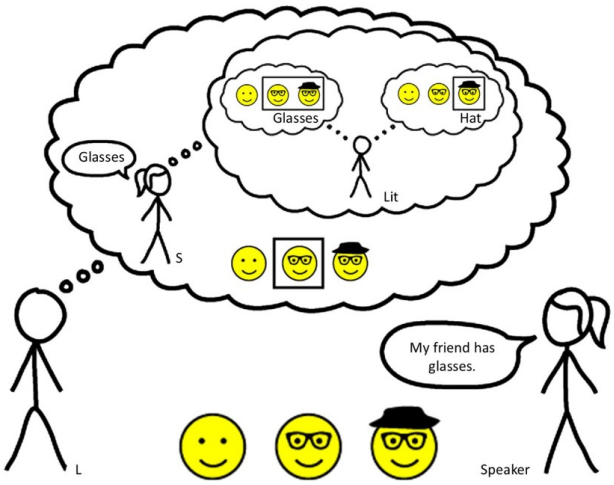
1.2 Rational Speech Act Models

The Rational Speech Act (RSA) model builds on the Bayesian model by additional restrictions on the likelihood term $P_S(utterance \mid referent)$. The RSA models assume that speakers are approximately rational when they choose which utterances to say among all possible alternative utterances. The RSA model views communication as recursive reasoning between a speaker and a listener, hypothesizing that a rational listener assumes the speaker is rational and has already chosen the best utterance that would allow a naive literal listener, L_0, to correctly infer the referent given the literal meaning of the utterance. Therefore, there are three levels of inference in communication in the context of the RSA model: **Rational listener**, L, reasons about **rational speaker**, S, to infer the utterance, and in L's reasoning, S also reasons about **literal listener**, L_0, to choose the best utterance.

One simple example demonstrating how RSA works is a referent game scenario shown in Figure 1 adopted from (Goodman & Frank, 2016). In this referent guessing game, a speaker and a listener can both see three faces with name N (none), G (glasses), and HG (Hat and glasses), from left to right respectively. When the speaker says "my friend has glasses", the listener needs to guess which one is the speaker's friend. The RSA model hypothesizes that the listeners share the intuition that the sentence 'my friend has glasses' refers to G and not HG, proposing that listeners reason about the speakers' mental state, and realize that if the speakers meant to refer to the one with hat and glasses (HG), they could have said "my friend has a hat" to avoid any ambiguity. The fact that they did not

say so implied HG was not the referent.

Figure 1
Reference game: an example demonstrating how RSA works.



Example above shows that the implicatures of the listeners can be formed not only on the utterance that was generated by the speakers, but also on possible alternatives that were not used by the speakers. Similarly, in the context of pronoun interpretation, it is also important to find all alternative pronouns and all possible interpretation of each pronoun in order to build a complete RSA model. A rational listener in the RSA model first finds all possible referents for a pronoun and then asks themselves that are there alternative pronouns for the speaker that can avoid ambiguities and why the speaker did not use these pronouns. The fact that they did not use alternative pronouns gives some hints as to who is the referent. This reasoning process can be captured by a system of probabilistic formulas which is adopted from Schulz et al. (2021). The rest of this section introduces each formula in turn.

1.2.1 Literal Listener

The listener's interpretation process assumes that the speaker is obeying the maxim of quality. The speaker only gives true and grammatical information. At the base of the recursive reasoning, the literal listener L_0 interprets an utterance literally without any

preferences. The literal listener's interpretation is captured in the Equation (6).

(6) Literal Listener:

For all $utterance \in all\ possible\ pronouns$ and $referent \in all\ possible\ referents$,

$P_{L_0}(referent \mid utterance) = \frac{P(referent \cap utterance)}{P(utterance)}$

This calculation is able to exclude all the referents that are not in the meaning of the utterance by assigning 0 possibility for ungrammatical utterance-referent pairs. After excluding all ungrammatical pairs, the literal listener then normalizes the prior world knowledge, $P(referent)$, for all grammatical utterance-referent pairs to form the literal interpretation of a pronoun. For example, if there are three possible referents — a, b, and c — with prior probabilities of 0.1, 0.6, and 0.3, respectively. Now if the spoken pronoun cannot refer to the referent b because of grammar rules in that language, the literal listener first assigns 0 to referent b for literal interpretation, and then normalizes other possible referents. So literal listener's interpretations are $P_{L_0}(b \mid pronoun) = 0$, $P_{L_0}(a \mid pronoun) = \frac{0.1}{(0.1+0+0.3)} = 0.25$, and $P_{L_0}(c \mid pronoun) = \frac{0.3}{(0.1+0+0.3)} = 0.75$.

1.2.2 Rational Speaker

The main characteristic of RSA models is that it thinks about a rational speaker who considers the literal listener and maximizes the utility to generate an utterance.

Speaker Utility. The speaker utility in Equation (7) formalizes the maxim of Quantity, which is partially determined by the informativeness of the message. Frank and Goodman (2012) proposes to take the natural log of the literal listener interpretation probability ($P_{L_0}(referent \mid utterance)$) added by the costs of the message to represent the speaker utility. The natural log of the likelihood is called negative surprisal which shows the informativity of the message to convey the world. A high likelihood will have a low negative surprisal — it is not surprising at all to see it happen. The frequencies from the corpus study are integrated into the model by being encoded as part of the cost function in

Equation (8): Schulz et al. (2021) takes the natural logarithm of the proportions in the corpus as the cost. For example, if the relative frequency of a pronoun compared to its alternatives in a corpus is 0.47, then its cost is $log(0.47) = -0.328$. The absolute value of cost of a pronoun with a higher frequency is lower than the absolute value of cost of a pronoun with a lower frequency. The negative cost is added to the utility function. According to RSA model, a rational speaker wants to maximize the utility by a trade-off between minimizing the surprisal of referent given an utterance for the literal listener and maximizing the frequency of the frequency.

(7) Speaker Utility:

$$U_S(utterance,\ referent) = log(P_{L_0}(referent \mid utterance)) + C(utterance)$$

(8) Cost Function:

$$C(utterance) = \log(P(utterance))$$

Speaker Decision-making. The speaker's probability of using a message to express world is given by the Softmax choice rule which assumed that speakers are approximately rational and they select utterance with the guidance of speaker utility $U_s(m, w)$ in Equation (9); meanwhile, there may be some variability in the mental computations because of time and resource limitations, which is why the Softmax choice rule contains a parameter α. The term $\alpha > 0$ controls the speaker's rationality in choosing utterances. An α that is greater than or equals to 1 means a speaker fully maximizes the speaker utility or even overuses the utility. An α that is between 0 and 1 means a speaker may not always choose the most useful utterance subject to time and resource limitations. Schulz et al. (2021) uses Bayesian parameter estimation with Markov-chain Monte Carlo methods implemented in WebPPL to estimate the optimal value for α is 0.93. Therefore, I adopts $\alpha = 0.93$ in this study.

(9) Speaker Decision-making (Softmax Choice):

$P_S(utterance \mid referent) \propto exp(\alpha \times U_S(utterance,\ referent))$

1.2.3 Rational Listener

The listener's rational interpretation, integrating the prior world knowledge $P(referent)$ and the likelihood from a rational speaker $P_S(utterance \mid referent)$, is captured by Bayesian inference as Equation (4) in the Bayesian model. The full RSA model is shown in (10) below. The RSA Model predicts that if independent estimates of the prior $P(referent)$, utterance frequency $P(utterance)$, and posterior $P_L(referent \mid utterance)$ probabilities were obtained, the system of equations in (10) would approximately hold.

(10) a. Rational Listener:

$P_L(referent \mid utterance) \propto P(referent) \times P_S(utterance \mid referent)$

b. Speaker Decision-making (Softmax Choice):

$P_S(utterance \mid referent) \propto exp(\alpha \times U_S(utterance,\ referent))$

c. Speaker Utility:

$U_S(utterance,\ referent) = log(P(referent \mid utterance) + C(utterance)$

d. Literal Listener:

For all $utterance \in pronouns$ and $referent \in referents$,

$P_{L_0}(referent \mid utterance) = \frac{P(referent \cap utterance)}{P(utterance)}$

e. Cost Function:

$C(utterance) = \log(P(utterance))$

Schulz et al. (2021) looks at the pronoun interpretation differences in English and French in utterances of the ambiguous finite sentence structure like the sentence (2) in the first puzzle (how and why interpretation varies across languages). The alternative utterance is a non-finite constructions such as "The postman met the streetsweeper before going home". Schulz et al. (2021) conducts a corpus study on both ambiguous finite

constructions and unambiguous non-finite constructions in order to find out the frequency of utterances, modeled by $P(utterance)$, so that they know the cost of the utterances, $Cost(utterance)$. They assume that prior beliefs with $P(referent = postman) = 0.8$ and $P(referent = streetsweeper) = 0.2$ based on data from previous studies. The alpha value is set to be 0.93. In addition, they design an experiment to estimate the actual posterior, modeled by $P_L(referent \mid utterance)$. Their study contains a gap completion task as shown in (11) with an aim to find out listeners' interpretation of the pronoun. After getting all parameters in the RSA model, they are able to generate probabilistic predictions. If the RSA model is correct, the resolution predictions should correlated with the actual interpretation biases that are estimated directly the experiment. The information of the parameters and a comparison between the RSA model prediction and actual interpretation biases is shown in Table 2. Schulz et al. (2021) shows that the experimental data has a very good fit of the RSA model predictions for the differences in antecedent choices in pronoun interpretation between French and English.

(11) The postman$_i$ met the streetsweeper$_j$ before **he**$_{i/j}$ went home.

_____ went home.

Table 2

RSA Model Prediction and Actual Biases example

	English	French
Costs	C(finite) = –0.117	C(finite) = –1.309
	C(non-finite) = –2.207	C(non-finite) = –0.315
Prior beliefs	P(postman) = 0.8	P(postman) = 0.8
	P(streetsweeper) = 0.2	P(streetsweeper) = 0.2
α	0.93	0.93
Model Predictions	$P(postman\|finite) = 23\%$	$P(postman\|finite) = 51\%$
	$P(streetsweeper\|finite) = 77\%$	$P(streetsweeper\|finite) = 49\%$
Actual Biases	$P(postman\|finite) = 20\%$	$P(postman\|finite) = 62\%$
	$P(streetsweeper\|finite) = 80\%$	$P(streetsweeper\|finite) = 38\%$

1.3 Present Study

In order to further explore the ability of probabilistic models — Bayesian models and RSA models — to explain cross-linguistic variation in pronoun interpretation preferences, this study looks at a different language — Mandarin Chinese. Mandarin is also able to represent the pronoun interpretation challenge with a special pronoun — reflexive pronoun *ziji*. The binding possibilities for the Chinese Mandarin reflexive pronoun *ziji* "self" are shown in (12) below. The reference of the reflexive *ziji* "self" is ambiguous, permitting both a local and a non-local construal represented by the fact that *ziji* is coindexed with both *Zhang Wei* (with the subscript i) and *Xiao Ming* (with the subscript j). Meanwhile, this study also investigates the second puzzle — the interpretation variation across context — by attempting to apply the models to account for the different interpretation in different context. Specifically, I will look at item-level variation when building Bayesian and RSA models instead of consistently using a uniform prior and likelihood probabilities for all items as is typically done in experimental studies on pronoun interpretation. In short, the present study tests the Bayesian models and RSA models in a different language (Mandarin Chinese) with a novel set of pronouns (reflexive pronouns), predicting item-level interpretation variation.

(12) [张 伟]$_i$ 说 [小 明]$_j$ 把 [自己]$_{i/j}$ 弄糊涂了。
[Zhang Wei]$_i$ says [Xiao Ming]$_j$ BA [ziji]$_{i/j}$ confuse$_{PERF}$.
'[Zhang Wei]$_i$ says that [Xiao Ming]$_j$ confused [self]$_{i/j}$.'

A **World Knowledge Bias Test** is conducted to find out people's prior biases for each stimuli as an estimate for the prior $P(referent)$. A **Pronoun Production Task** is also conducted to find out what pronoun do people prefer to use to refer to a specific antecedent and the result of the task is used as an estimate for the likelihood $P_S(utterance \mid referent)$. In order to evaluate the Bayesian model, a **Pronoun Interpretation Task** is conducted which tried to find out how do people interpret

pronouns in Mandarin Chinese. This experiment generates actual interpretation bias, the posterior $P_L(referent \mid utterance)$. If Bayesian models successfully fit the experimental data well, we may have a direction for understanding listeners' reasoning when interpreting an utterance — a listener's interpretation bias relies jointly on the prior knowledge that a particular referent will be more likely to be mentioned than other alternatives under a specific context and on the likelihood that the speaker would have chosen a specific pronoun to convey the information of that referent.

For RSA models, we use results from the Pronoun Interpretation Task and World Knowledge Bias Test mentioned above as actual interpretation biases and as an estimate for the prior $P(referent)$. Instead of the Pronoun Production Task, we conduct a **corpus study** to find out the frequency of each possible utterance so that we can get the cost of utterances, $C(utterance)$. If RSA models successfully fit experimental data, in addition to the joint factors mentioned in last paragraph, we may also understand listeners' reasoning in communication as recursive reasoning between rational listeners and rational speakers.

The rest of the paper is organized as follows. The grammatical usages of *ziji* and all other possible alternative pronouns are introduced in the next section. I then discuss in detail three experiments and a corpus study. Next I fit the model predictions with experimental data and generates results. Last but not least, it provides a general discussion of results.

2 Pronouns in Mandarin Chinese

In generative grammar, there are three kinds of NP-types pronouns: reflexive pronouns, personal pronouns, and Referential-expressions (R-expressions). In Mandarin Chinese, reflexive expressions like 自己 *ziji* 'self' and combinations of pronouns + *ziji* like 他自己 *taziji* 'himself', 他们自己 *tamenziji* 'themselves', etc. are reflexive pronouns since they must be associated with their antecedents. (13) and (14) is an example of reflexive expressions in Mandarin Chinese. Personal pronouns like 他/她 *ta* 'he/she' may be used with or without antecedents. (22) is an example of personal pronouns in Mandarin Chinese. Other noun phrases (names, *zhe-ge xiaohai* 'this child', etc.) are R-expressions, which do not require an antecedent (Huang, Li, & Li, 2009).

Reflexive pronouns in Mandarin Chinese may take a monomorphemic form, 自己 *ziji* 'self', or a compound form which combines *ziji* with a personal pronoun such as 我自己 *woziji* 'myself'. (13) shows that the behavior of the compound reflexive in Mandarin Chinese is similar to its equivalent in English, represented by the fact that *taziji* and *Xiao Ming* are coindexed with the subscript j in (13). It cannot take the non-local noun phrase (NP) *Zhang Wei* as its antecedent, nor can it exist without an antecedent. In Mandarin Chinese, three third-person singular reflexive pronouns "himself, herself, and itself" are all pronounced as "*taziji*" .

(13) [张 伟]$_i$ 说 [小 明]$_j$ 把 [他自己]$_j$ 弄糊涂了。
[Zhang Wei]$_i$ says [Xiao Ming]$_j$ BA [taziji]$_j$ confuse$_{PERF}$.
'[Zhang Wei]$_i$ says that [Xiao Ming]$_j$ confused [himself]$_j$.'

However, the bare reflexive 自己 *ziji* has a different set of possible interpretations in contrast to (13). The sentence in Example (14) is ambiguous, permitting both a local and a non-local construal of *ziji*, represented by the fact that *ziji* is coindexed with both *Zhang Wei* (with the subscript i) and *Xiao Ming* (with the subscript j). The lack of overt morphological features in *ziji* allows it to take any antecedents regardless of the gender or

number.

(14) [张 伟]$_i$ 说 [小 明]$_j$ 把 [自己]$_{i/j}$ 弄糊涂了。
[Zhang Wei]$_i$ says [Xiao Ming]$_j$ BA [ziji]$_{i/j}$ confuse$_{PERF}$.
'[Zhang Wei]$_i$ says that [Xiao Ming]$_j$ confused [self]$_{i/j}$.'

Moreover, personal pronouns act differently than reflexive pronouns. In contrast with (13), replacement of *taziji* with the pronoun 他 *ta* gives an opposite pattern of interpretive possibilities as shown in Example (22), where *ta* may refer to the non-local NP (with the subscript i) or others (with the subscript k) but not the local-NP (with the subscript j). Only when the antecedent does not c-commend or bind the personal pronoun *ta*, it may refer to the local NP, as shown in Example (16), where the sentence has a possessive pronoun *ta-DE* by adding a suffix *DE* to the personal pronoun. In this case, *ta* can be coindexed with *Zhang Wei*, *Xiao Ming*, or others (Huang et al., 2009). In Mandarin Chinese, all third-person singular personal pronouns are all pronounced as "*ta*" but with different characters — 他 "he/him", 她 "she/her", and 它 "it".

(15) [张 伟]$_i$ 说 [小 明]$_j$ 把 [他]$_{i/*j/k}$ 弄糊涂了。
[Zhang Wei]$_i$ says [Xiao Ming]$_j$ BA [ta]$_{i/*j/k}$ confuse$_{PERF}$.
'[Zhang Wei]$_i$ says that [Xiao Ming]$_j$ confused [him]$_{i/*j/k}$.'

(16) [张 伟]$_i$ 说 [小 明]$_j$ 把 [他]$_{i/j/k}$ 的 工作 做完了。
[Zhang Wei]$_i$ says [Xiao Ming]$_j$ BA [ta]$_{i/j/k}$ DE work finish$_{PERF}$.
'[Zhang Wei]$_i$ says that [Xiao Ming]$_j$ finished [his]$_{i/j/k}$ work.'

Pronouns in Mandarin Chinese are interesting in the way that the use of the reflexive *ziji* is special and there are more alternative pronouns for different meanings in Mandarin than English does, which means that the process of pronoun resolution is more complicated — there are multiple ways for listeners to interpret *ziji* while for each different interpretation there are also multiple alternative pronouns for speakers to deliver the same referent information. Would the additional choices the speaker has impact listener

preferences and is RSA model capable of explaining pronominal interpretation preferences in a more complicated system? That is why this study chooses to use pronoun resolution in Mandarin Chinese as a test case to the Bayesian and RSA model.

3 Methods

The goal of this search is to test the Bayesian model and RSA model on empirical data. To reach the goal, it is important to know what kind of empirical data is needed. The Bayesian model is shown in Equations (4). The full version of RSA model is shown in Equations (10). For the Bayesian model, we need empirical data for the posterior $P_L(referent \mid utterance)$, the likelihood $P_S(utterance \mid referent)$, and the prior $P(referent)$. As the Bayesian model, the RSA model also needs empirical data for the posterior and the prior. However, since the likelihood can be calculated from the the combination of pronoun frequency and grammatical rules in the RSA model, the empirical data for likelihood is not needed, but instead, I need the frequency for each pronoun $P(utterance)$.

Three experiments and a corpus study are designed accordingly. A **Pronoun Interpretation Task** is conducted for the posterior $P_L(referent \mid utterance)$. A **World Knowledge Bias Test** is conducted for the prior $P(referent)$. The empirical measurements of the likelihood distribution $P_S(utterance \mid referent)$ is from a **Pronoun Production Task**. The frequency of the pronoun $P(utterance)$ is from a **Corpus Study**. All three online experiment were designed on PennController for Internet Based Experiments[1] , an open-science platform for online experiments (Schwarz & Zehr, 2018). Each experiment lasts about 10-15 minutes. Each experiment has training and quality control trials. All participants were recruited on different Chinese social media, such as Douban and Weibo. Three experiments share the same entry link and the system randomly selects one task for people who click the entry link. A person can only take one of the three

[1] The code for three experiments is available in my GitHub: `https://github.com/zhaofylisa/Pronoun-Resolution-in-Mandarin`.

experiment and cannot take the experiment repeatedly. In general, 97% of the participants claimed that they speak Mandarin as their only native language. 67% of the participants are from Beijing, Tianjin, and Hebei where the potential side factor of varieties of Chinese is limited.

This section will first present the stimuli that are used in all three experiments and will introduce each task in turn. Before conducting experiments, as a human subjects researcher I have completed courses and a certification on the Collaborative Institutional Training Initiative Program in order to be engaged in research involving human subjects like the present study.

3.1 Stimuli

30 root stimuli with sentence structure either in (17) or (18) were created. In order to create variety of stimuli, 15 of the root stimuli are in **coargument condition**, which use pronouns *ziji*, *ta*, and *taziji*. The other 15 stimuli are in **possessor condition**. To make a pronoun possessive in Mandarin, a suffix *DE* is added to it, just like English possessive suffix 's after nouns. For example, *ta-DE* is equivalent to English "his". The sentence structure for coargument condition is in (17) and that of possessor condition is in (18).

The sentence structures show that each stimuli includes a main clause with "says" as the main verb and a subordinate that-clause after "says". The subject of the main clause is called Non-local Noun Phrase, **Non-local NP**, and the subject of the subordinate clause is called Local Noun Phrase, **Local NP**. For example, in sentence "Lisa says that Erin likes her", the whole sentence is a main clause and "that Erin likes her" is a subordinate clause, such that Lisa is the Non-local NP and Erin is the Local NP.

(17) [Non-local NP]$_i$ 说 [Local NP]$_j$ 把 [PRONOUN] [VERB$_{PERF}$]。
[Non-local NP]$_i$ says [Local NP]$_j$ BA [PRONOUN] [VERB$_{PERF}$].
'[Non-local NP]$_i$ says that [Local NP]$_j$ [VERB$_{PERF}$] [PRONOUN].'

(18) [Non-local NP]$_i$ 说 [Local NP]$_j$ 把 [PRONOUN] 的 [Object NP] [VERB$_{PERF}$]。
[Non-local NP]$_i$ says [Local NP]$_j$ BA [PRONOUN] DE [Object NP] [VERB$_{PERF}$].
'[Non-local NP]$_i$ says that [Local NP]$_j$ [VERB$_{PERF}$] [PRONOUN]'s [Object NP].'

In order to keep the existence of ambiguity and to avoid people interpreting a pronoun based on the gender of the Non-local NP and the Local NP, this study decides to only use male noun phrases and male pronouns (i.e. he/him and himself). Both non-local NP and local NP in the stimuli are randomly chosen from four placeholder names commonly used for anonymous males in China: 小明 *Xiao Ming*, 李华 *Li Hua*, 王刚 *Wang Gang*, and 张伟 *Zhang Wei*. Each name is associated with a cartoon headshot shown in Figure 2. The verbs in the subordinate clauses are chosen within the range of positive, negative, and neutral sentiment words. All the verbs are in the perfect aspect. Additionally, all stimuli use *BA*-construction. The *BA*-construction was used to: (1) avoid anaphoric object drop. An object can be dropped when there is an antecedent in Chinese Mandarin. Anaphoric object drop can be avoided in *BA*-construction since an object is required after *BA* in Mandarin syntax; (2) avoid having the target pronoun in the final position of the sentence with extra distractions since *BA*-construction changes the word order from Subject-Verb-Object to Subject-*BA*-Object-Verb.

Figure 2
Headshots for four characters

(i) *Xiao Ming* (ii) *Li Hua* (iii) *Wang Gang* (iv) *Zhang Wei*

In addition to the experimental stimuli, five quality control trials were designed which were unambiguous to make sure participants fully read sentences and to ensure the

quality of online experiment. All experiment will share same set of root stimuli but in different formats depending on the task of each experiment. The specific format of the stimuli will be specified as each experiment is presented.

3.2 Experiment 1: Pronoun Interpretation Task

Does the listener have any preference when resolving ambiguous pronouns like *ziji* and *ta* in Mandarin? The goal for this pronoun interpretation task it to find out how do people interpret different pronouns in Mandarin. That listeners have a preference is a premise to this project since if there is no preference we can just use a uniform probability model — a model in which every outcome has equal probability — to explain the pronoun interpretation process. In other words, there is nothing interesting to explain in this case. The result of this pronoun interpretation task is used as an estimate of $P_L(referent \mid utterance)$ in both Bayesian and RSA models.

3.2.1 Participants

147 participants were recruited on Chinese social media in the experiment. 7 participants were removed prior to analysis due to low accuracy in the quality control trials. So data from in total 135 valid participants were collected. All participants were native Mandarin Chinese speakers from mainland China with age range from 18 to 40. The experiment lasted approximately 10 - 15 min. They were paid 12 Chinese Yuan (about 2 dollars) for their participation in the experiment. All participants gave informed consent by reading and submitting an online consent form before the experiment.

3.2.2 Procedure

The design of the experiment is as the following. Given a full sentence with sentence structure shown in (20) and a passive question asking who the pronoun refers to (for example, “Who is confused?”), participants were asked to answer the question by clicking the picture of the character as fast as possible. There are three possible options: the

character of Non-local NP, the character of Local NP, and the character of Others. An example of the task is given in Figure 3, where the first line in black color is the stimuli; the second line in red color is the question asking who the pronoun in the stimuli refers to; the third line in blue color gives participant instruction to answer the question by clicking one of the three characters below. In order to avoid bias from the provided options, participants saw the full sentence and the questions first and had 3 seconds to decide who the pronoun refers to without seeing any options and the order of options is shuffled with each trial. They could skip to see the options by pressing spacebar on keyboard if they were able to quickly choose the referent in less than 3 seconds.

Figure 3

Example of the Pronoun Interpretation Task

Informed consent was obtained from each participant before the experiment began. Then they were asked to provide demographic information such as age, gender, native language, and their parents' native language. Next, the experiment instruction was shown on the screen. Then, participants were then given five examples of the task as practice trials before beginning the experiment. The stimuli used in the practice trials were unambiguous so there is correct referent for a pronoun. An example of the stimuli used in

practice trials is shown in (19). The practice trials can help participants get familiar with the task procedure. A debrief form and a payment instruction form were presented to participants after the experiment.

(19) [张 伟]$_i$. 把 [自己]$_i$ 打了。
[Zhang Wei]$_i$ BA [ziji]$_i$ [hit$_{PERF}$].
'Zhang Wei$_i$ hit himself$_i$.'

3.2.3 Stimuli

There are in total thirty stimuli. Fifteen of them are in coargument condition with sentence structure in (17), and the other fifteen of them are in possessor condition with sentence structure in (18). For each sentence, there are three types for the pronouns — ziji, taziji, and ta. Therefore there are $30 \times 3 = 90$ stimuli in total. The permutation of two sentence types and three pronounce types yields six experimental conditions. These conditions are summarized in (20). Five quality control stimuli with three types of pronouns were used which were unambiguous to ensure the quality of online experiment. The experimental items and quality control items are distributed into three lists in a pseudo-Latin square fashion. In other words, each participant sees each root stimuli only once in the experiment, rather than seeing repeated root stimuli in all conditions. The condition of pronoun type for each root stimuli is randomly selected.

(20) 1. coargument + *ziji*:

(i) [张 伟]$_i$ 说 [小 明]$_j$ 把 [自己]$_{i/j}$ 弄糊涂了。
[Zhang Wei]$_i$ says [Xiao Ming]$_j$ BA [ziji]$_{i/j}$ confuse$_{PERF}$.
'[Zhang Wei]$_i$ says that [Xiao Ming]$_j$ confused [self]$_{i/j}$.'

2. coargument + *taziji*:

(i) [张 伟]$_i$ 说 [小 明]$_j$ 把 [他自己]$_j$ 弄糊涂了。
[Zhang Wei]$_i$ says [Xiao Ming]$_j$ BA [taziji]$_j$ confuse$_{PERF}$.
'[Zhang Wei]$_i$ says that [Xiao Ming]$_j$ confused [himself]$_j$.'

3. coargument + *ta*:

   (i) [张 伟]$_i$ 说 [小 明]$_j$ 把 [他]$_{i/*j/k}$ 弄糊涂了。
   [Zhang Wei]$_i$ says [Xiao Ming]$_j$ BA [ta]$_{i/*j/k}$ confuse$_{PERF}$.
   '[Zhang Wei]$_i$ says that [Xiao Ming]$_j$ confused [him]$_{i/*j/k}$.'

4. possessor + *ziji*:

   (i) [张 伟]$_i$ 说 [小 明]$_j$ 把 [自己]$_{i/j}$ 的 工作 做完了。
   [Zhang Wei]$_i$ says [Xiao Ming]$_j$ BA [ziji]$_{i/j}$ DE work finish$_{PERF}$.
   '[Zhang Wei]$_i$ says that [Xiao Ming]$_j$ finished [ziji]$_{i/j}$ work.'

5. possessor + *taziji*:

   (i) [张 伟]$_i$ 说 [小 明]$_j$ 把 [他自己]$_j$ 的 工作 做完了。
   [Zhang Wei]$_i$ says [Xiao Ming]$_j$ BA [taziji]$_j$ DE work finish$_{PERF}$.
   '[Zhang Wei]$_i$ says that [Xiao Ming]$_j$ finished work of [himself]$_j$.'

6. possessor + *ta*:

   (i) [张 伟]$_i$ 说 [小 明]$_j$ 把 [他]$_{i/j/k}$ 的 工作 做完了。
   [Zhang Wei]$_i$ says [Xiao Ming]$_j$ BA [ta]$_{i/j/k}$ DE work finish$_{PERF}$.
   '[Zhang Wei]$_i$ says that [Xiao Ming]$_j$ finished [his]$_{i/j/k}$ work.'

3.2.4 Result and Discussion

A general result for this Pronoun Interpretation task for six experimental conditions is shown in Fig 4. Paired samples t-tests were performed to examine apparent differences between conditions for each pronoun. For the ambiguous pronoun *ziji*, Fig 4(i) shows that there is a significant preference for Local NP over Non-local NP for possessor + *ziji* condition ($t = 6.15$, $p < 0.05$, $df = 28$) while there is no significant preference between Non-local NP and Local NP for coargument+*ziji* condition ($t = -0.422$, $p = 0.676$, $df = 28$). For the unambiguous pronoun *taziji*, Fig 4(ii) shows that there are significant preferences for Local NP over Non-local NP in both coargument +*taziji* and possessor + *taziji* ($t = 19.066$, $p < 0.05$, $df = 28$ and $t = -24.894$, $p < 0.05$, $df = 28$, respectively). Under the coargument+*ta* condition, there is an apparent preference for Non-local NP interpretation over both Local NP ($t = 33.129$, $p < 0.05$, $df = 28$) and Others ($t = 37.729$, $p < 0.05$, $df = 28$) interpretations. Others interpretation is the least preferred ($t = -3.803$,

Figure 4

Results for Pronoun Interpretation Task (overall)

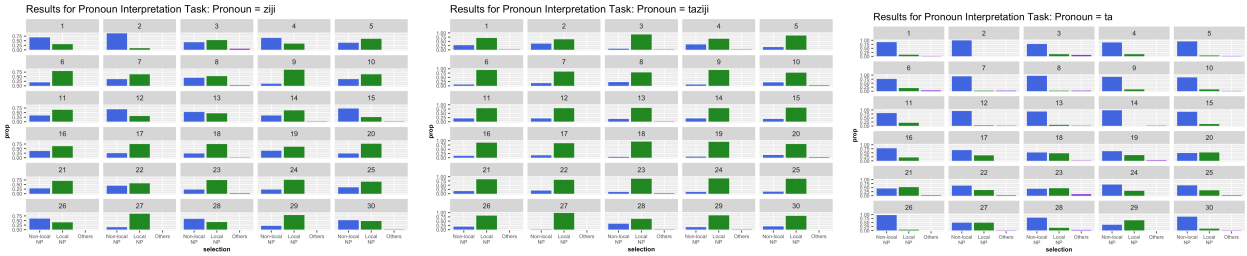
(i) *pronoun = ziji* (ii) *pronoun = taziji* (iii) *pronoun = ta*

p < 0.05, df = 28, compared to Local NP interpretation). Under the possessor + *ta* condition, there is also an apparent preference for Non-local NP interpretation over both Local NP (t = 4.375, p < 0.05, df = 28) and Others (t = 10.174, p < 0.05, df = 28) interpretations. Others interpretation is still the least preferred (t = -5.674, p < 0.05, df = 28, compared to Local NP interpretation).

Item-level result is shown in Fig 5. Item 1 through item 15 are in coargument condition, while item 16 through item 30 are in possessor condition. Although we got a general result from statistical tests mentioned above, we cannot ignore the individual differences across items. For example, when the pronoun is *ziji*, participants have differences in interpretation preference between items 1 and 8: for item 1 (the verb is confuse), participants prefer the Non-local NP as the referent of *ziji* to the Local NP, whereas for item 8 (the verb is trip-over), the opposite is true. In addition, it is possible that the direction of the preference may be the same while the magnitude of the preference still varies. For example, both item 1 and item 2 (the verb is betray) have Non-local NP preference when pronoun is *ziji*, but item 2 has stronger preference for Non-local NP preference than item 1.

Figure 5

Results for Pronoun Interpretation Task (by item)



(i) *pronoun = ziji* (ii) *pronoun = taziji* (iii) *pronoun = ta*

The differences in interpretation preference and in extent of the preference show us that our experimental data are not directional and qualitative but graded and quantitative. The Bayesian model and RSA model as probabilistic approaches to pragmatics can be an attempt to capture the fine item-level variation in our pronoun interpretation task.

3.3 Experiment 2 - World Knowledge Bias Test

Who is most likely to be the object of an action in situation like (21), the non-local NP *Zhang Wei*, the local NP *Xiao Ming*, or another person? The goal of this World Knowledge Bias Test is to find out people's prior world knowledge bias. The result of the prior world knowledge bias test tells us the referent that is more likely to be mentioned next in a specific communication scenario regardless of the referring expression used. The referent is independent of how it is linguistically encoded. The result of this task is used as an estimate of $P(referent)$ in both Bayesian and RSA models.

(21)

: “

把 [?] 弄糊涂了 。
Zhang Wei: “Xiao Ming BA [?] confuse$_{PERF}$.
‘Zhang Wei says, “Xiao Ming confused [?].” ’

3.3.1 Participants

33 participants were recruited on Chinese social media. 5 participants were removed prior to analysis due to low accuracy in the quality control trials. Therefore experimental

data from 28 valid participants in total were collected. All participants were native Mandarin Chinese speakers from mainland China with age range from 18 to 40. The experiment lasted approximately 10 - 15 min. They were paid 12 Chinese Yuan (about 2 dollars) for their participation in the experiment. All participants gave informed consent by reading and submitting an online consent form before the experiment.

3.3.2 Procedure

Participants were given a real-world situation with a question mark in the target pronoun position, and they needed to choose a character that is most likely to be the object of the verb. One example of the task is shown in Figure 6, where the real-world situation is at the top and participants needed to choose the characters between *Zhang Wei*, *Xiao Ming*, or Others to fill in the blank. The pictures of the character were used instead of their names because I wanted to inform participants this is not a sentence but a real-world situation, so that they would not try to pick an option that makes the sentence grammatical. In order to avoid bias from provided options, participants had 3 seconds thinking time before seeing three options and the order of options is shuffled with each trial.

Informed consent was obtained from each participant before the experiment began. They were also asked to provide demographic information such as age, gender, native language, and their parents' native language. At the end of the experiment, a debrief form and a payment instruction form were presented to participants. Before starting the task, the experiment instruction was shown on the screen. Participants were then given one examples of the task as a practice trial before beginning the experiment. Since there is no single correct answer for this kind of task, the practice trial is more like an interactive example to help participants get familiar with the task. In the practice trial, participants saw a situation in (22) and three options are Zhang Wei, Xiao Ming, and Others.

Figure 6

Example of the World Knowledge Bias Test

(22) : “ 把 [?] 的 杯子 打碎了。”
Zhang Wei: “Xiao Ming BA [?] DE cup break$_{PERF}$.”
‘Zhang Wei says, “Xiao Ming broke the cup of [?].”’

Participants were instructed to click three options one by one to simulate the reasoning behind each choice. The three options correspond to different intuitive scenarios. When participants click one option, the corresponding scenario below is shown at the same time as an example to explain how to reason about the task. Only after checking all options, participants could officially start the experiment.

- Scenario 1: Someone asked Zhang Wei why he brought a cup, and Zhang Wei explained, “**Xiao Ming broke my cup**. I’m going to the supermarket to buy a new one.” => Choose **Zhang Wei**

- Scenario 2: Someone asked Zhang Wei why he gave Xiao Ming a cup as gift, Zhang Wei explained, "**Xiao Ming broke his own cup**. So I give him a new one as gift." => Choose **Xiao Ming**

- Scenario 3: When Zhang Wei and his family were having tea at home, Zhang Wei has a cup but a family member did not find their cup, Zhang Wei explained, "**Xiao Ming broke your cup**. I'll go get you another one." => Choose **Others**

3.3.3 Stimuli

There are in total thirty stimuli, as introduced in Section 3.1. Fifteen of them are in coargument condition with structure in (23), and the other fifteen of them are in possessor condition with structure in (24). The Non-local NP and Local NP in stimuli structures are pictures of characters instead of just names so that participants would not think the stimuli is a sentence. Different punctuation was used: a combination of colon and quotation marks in Chinese means someone is saying something; a question mark occurs right after *BA*, asking participants who is more likely to be in this position in a saying. An example of stimuli in coargument condition is shown in (21) and an example of stimuli in possessor condition is shown in (22).

(23) [Non-local NP]: "[Local NP] 把 [**?**] [VERB$_{PERF}$]。"
[Non-local NP]: "[Local NP] BA [**?**] [VERB$_{PERF}$]。"
'[Non-local NP] says, "[Local NP] [VERB$_{PERF}$] [**?**]." '

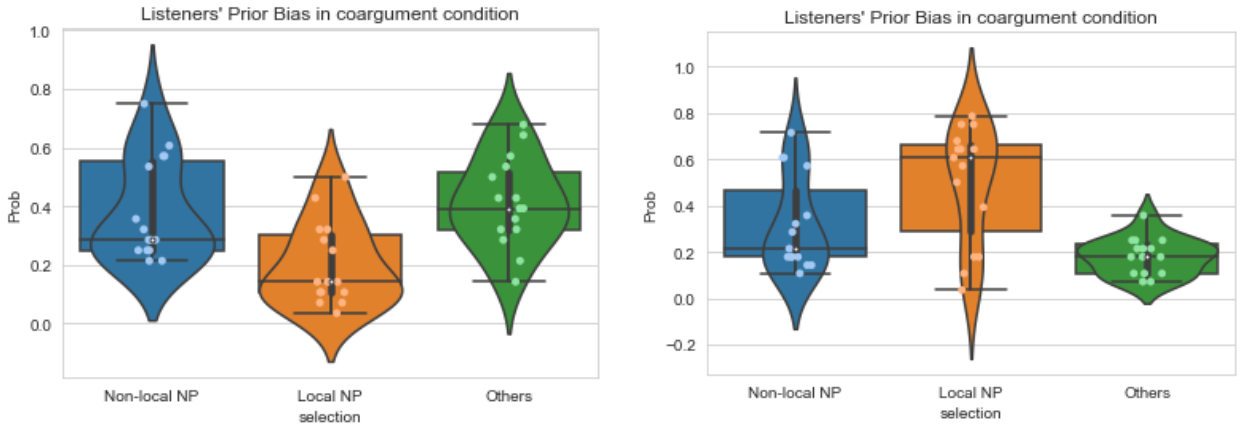
(24) [Non-local NP]: "[Local NP] 把 [**?**] 的 [Object NP] [VERB$_{PERF}$]。"
[Non-local NP]: "[Local NP] BA [**?**] DE [Object NP] [VERB$_{PERF}$]。"
'[Non-local NP] says that [Local NP] [VERB$_{PERF}$] [Object NP] of [**?**]." '

3.3.4 Result and Discussion

A general result for this World Knowledge Bias Test is shown in Fig 7. Paired samples t-tests were performed to examine apparent differences between participants'

Figure 7

Results for World Knowledge Bias Test (overall)



(i) *type = coargument* (ii) *type = possessor*

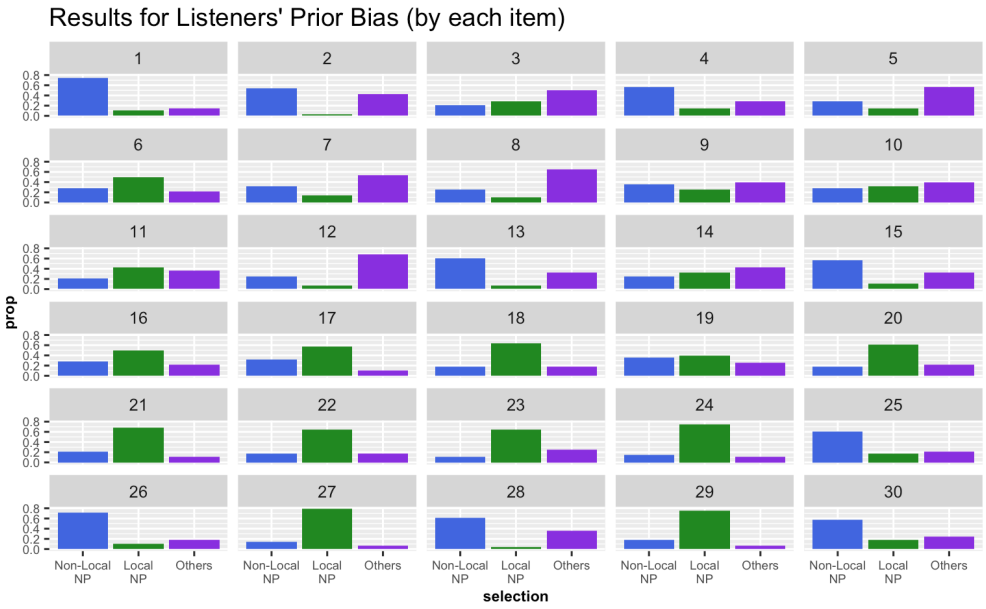
selections. In possessor condition, there exists a great bias for choosing the Local NP based on participants' world knowledge. The Local NP is the most likely object of an action in stimuli compared to Non-local NP and Others ($t = 2.12$, $p < 0.05$, $df = 28$ and $t = 4.57$, $p < 0.05$, $df = 28$, respectively). Meanwhile, there is a significant preference of selecting Non-local NP over selecting Others ($t = 2.39$, $p < 0.05$, $df = 28$). In coargument condition, the difference between P(Non-local NP) and P(Local NP) is significant ($t = 3.13$, $p < 0.05$, $df = 28$) and difference between P(Local NP) and P(Others) is significant too ($t = -3.97$, $p < 0.05$, $df = 28$). However, the difference between P(Non-local NP) and P(Others) is not significant ($t = -0.52$, $p > 0.05$, $df = 28$), which shows that participants do not have a general bias between Non-local NP and Others in terms of which person in the stimuli is more likely to be the object of an action.

The item-level result of the World Knowledge Bias Test is shown in Figure 8. The horizontal coordinate is participants' selection among three choices: Non-local NP, Local NP, and Others. The vertical coordinate is the possibilities of the selection. Similar to results in the first experiment, there are also many variation across items. For example, if you look at item 1, item 16, and item 8, you will find three different distribution of selection: for item 1 (verb is confuse), Non-local NP is the most likely object of the verb;

for item 16 (verb is send-somebody's-picture-to-social-media), Local NP is the most likely object of the specific action; for item 8 (the verb is trip-over), Others is the most likely object of the action among three possible choices. Additionally, the extent of the biases are also different across items. For example, item 1 (verb is confuse) has a strong Non-local NP preference than item 2 (verb is betray) does.

Figure 8

Results for World Knowledge Bias Test (by item)



3.4 Experiment 3 - Pronoun Production Task

The goal of the Pronoun Production Task is to find out what pronoun does a speaker prefer to use to refer to a specific antecedent in the context. The result is used as an estimate of $P_S(utterance \mid referent)$ in the Bayesian model.

3.4.1 Participants

79 participants were recruited on Chinese social media. 14 participants were removed prior to analysis due to low accuracy in the quality control trials. Therefore experimental data from 65 valid participants in total were collected. All participants were native Mandarin Chinese speakers from mainland China with age range from 18 to 40. The

experiment lasted approximately 10 - 15 min. They were paid 12 Chinese Yuan (about 2 dollars) for their participation in the experiment. All participants gave informed consent by reading and submitting an online consent form before the experiment.

3.4.2 Procedure

Given a sentence with a drop-down with different pronoun options in the target position and given a picture of the character that the pronoun should refer to, participants were asked to select the most natural pronoun from the drop-down menu to convey the character information. The four options are *ziji*, *taziji*, *ta*, and the name of the character such as Xiao Ming. Only the first three options are valid choices. The name of the character is not a pronoun and is not grammatical at any time in the context of this task so that it can serve as a quality control option to filter out participants who choose this option frequently. One example of the task is shown in Figure 9.

The text above the dotted line in the figure are instructions, saying that the picture of the character represents the missing person in the sentence and instructing participants to choose a pronoun that matches the person in the picture and complete the sentence according to their intuition and everyday usage of pronoun. One suggestion I gave the participants was that after choosing the pronouns, they could read through the sentences to see if they matched the pictures and if they sounded natural. The text below the dotted line is the stimuli with a drop-down and a character picture.

Informed consent was obtained from each participant before the experiment began. Then they were asked to provide demographic information such as age, gender, native language, and their parents' native language. Next, the experiment instruction was shown on the screen. Then, participants were then given five examples of the task as practice trials before beginning the experiment. The stimuli used in the practice trials were simpler than actual stimuli and since there is no single correct answer, there is no feedback about the correctness of their choices. The practice trials only help participants get familiar with

Figure 9
Example of the Pronoun Production Task

the task procedure. An example of the stimuli used in practice trials is shown in (25). The possible choices to this example can be *ziji* and *taziji*. A debrief form and a payment instruction form were presented to participants after the experiment.

(25) 张 伟 把 [DROP-DOWN] 打了。 +

Zhang Wei BA [DROP-DOWN] hit$_{PERF}$. + Picture of Zhang Wei
'Zhang Wei hit [DROP-DOWN].'

3.4.3 Stimuli

There are in total thirty stimuli, as introduced in Section 3.1. Fifteen of them are in coargument condition with structure in (17), and the other fifteen of them are in possessor

condition with structure in (18), but [PRONOUN] parts are replaced with a drop-down. In the drop-down, there are four possible options — *ziji*, *taziji*, *ta*, and the name of the given character. The order in the drop-down shuffles in each trial. For each root stimuli, there are two possible antecedent types as conditions, the non-local NP and the local NP. Is the picture of the given character the Non-local NP or the Local NP in the sentence? Technically, the task should also include Others as the antecedent. However, since we could not use reflexive pronouns to refer to other people, we assume the only possible pronoun participant could use for Others antecedent is *ta*. Therefore, there are $30 \times 2 = 60$ stimuli in total. The permutation of two sentence types and two conditions yields four experimental conditions. These conditions are summarized in (26). DD is the abbreviation for DROP-DOWN.

The experimental items are distributed into three lists in a pseudo-Latin square fashion. In other words, each participant will only see one condition of each stimuli. Each participant will either see the picture of the Non-local NP or the Local-NP but not both for the same root stimuli. The condition of pronoun type for each root stimuli is randomly selected for each participant. The experimental items and quality control items are distributed into three lists in a pseudo-Latin square fashion. In other words, each participant sees each root stimuli only once in the experiment, rather than seeing repeated root stimuli in all conditions. The condition of pronoun type for each root stimuli is randomly selected.

(26) 1. coargument + Non-local NP:

(i) [张 伟]$_i$ 说 [小 明]$_j$ 把 [DD] 弄糊涂了。
[Zhang Wei]$_i$ says [Xiao Ming]$_j$ BA [DD] confused.
'[Zhang Wei]$_i$ says that [Xiao Ming]$_j$ confused [DD].'

: Picture of Zhang Wei

2. coargument + Local NP:

   (i) [张 伟]$_i$ 说 [小 明]$_j$ 把 [DD] 弄糊涂了。
   [Zhang Wei]$_i$ says [Xiao Ming]$_j$ BA [DD] confused.
   '[Zhang Wei]$_i$ says that [Xiao Ming]$_j$ confused [DD].'

   : Picture of Xiao Ming

3. possessor + Non-local NP:

   (i) [张 伟]$_i$ 说 [小 明]$_j$ 把 [DD] 的 工作 做完了。
   [Zhang Wei]$_i$ says [Xiao Ming]$_j$ BA [DD] DE work finish$_{PERF}$.
   '[Zhang Wei]$_i$ says that [Xiao Ming]$_j$ finished work of [DD].'

   : Picture of Zhang Wei

4. possessor + Local NP:

   (i) [张 伟]$_i$ 说 [小 明]$_j$ 把 [DD] 的 工作 做完了。
   [Zhang Wei]$_i$ says [Xiao Ming]$_j$ BA [DD] DE work finish$_{PERF}$.
   '[Zhang Wei]$_i$ says that [Xiao Ming]$_j$ finished work of [DD].'

   : Picture of Xiao Ming

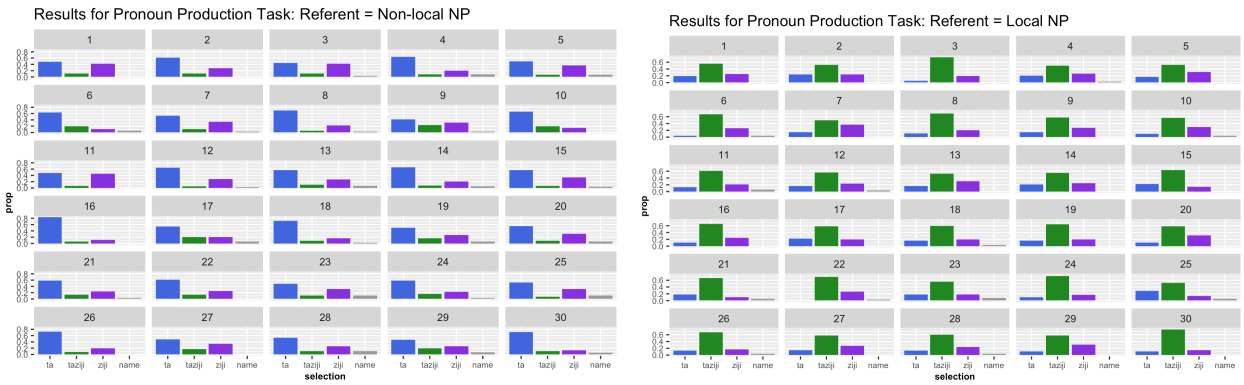
3.4.4 Result and Discussion

The item-level results for the Pronoun Production Task for different conditions are shown in Figure 10. The horizontal coordinate is participants' pronoun selection and the vertical coordinate is the probability for each selection. When the task directs participants to refer to the Local NP, there exists a clear preference to use *taziji* among all alternatives for all items. This experimental pattern is consistent with Mandarin grammar since *taziji* is an unambiguous pronoun which only can refer to the Local NP. When the task directs participants to refer to the Non-local NP, there exists a preference to use *ta* among all alternatives for all items. However, the extent of the preference varies across items. For item 1 (verb is confuse) and item 11 (verb is put-in-danger), the preference for *ta* is not as

strong as other items like item 16 (verb is send-somebody's-picture-to-social-media). Pronoun *ziji* is often used as well in item 1 and item 11. The result for pronoun production task is an estimate of $P_S(utterance \mid referent)$ in Bayesian model.

Figure 10

Results for Pronoun Production Task (by item)



(i) *referent = Non-local NP* (ii) *referent = Local NP*

3.5 Corpus Study

In the RSA model, the cost of an utterance is determined by the frequency. Therefore, a corpus study was conducted on the Chinese Web corpus 2017 Simplified (zhTenTen17) (Jakubíček, Kilgarriff, Kovář, Rychlý, & Suchomel, 2013). The corpus consists 16.5 billion tokens of Mandarin Chinese in simplified script, which were scraped from the Sketch Engine, a corpus manager and text analysis software (Kilgarriff et al., 2014). Since *BA* constructions was used in all stimuli, this corpus study searches through the corpus to count the frequency of the three target pronouns used after *BA* to get the frequency of the pronoun utterance ($P(utterance)$).

3.5.1 Result

The result of the corpus study is shown in Table 3. In coargument condition, *ta* is the most frequent pronoun among three alternatives in *BA*-construction, about 70.31%. The frequency of *ziji* in coargument condition in *BA*-construction is about 29.54%. The

frequency of *taziji* in coargument condition in *BA*-construction is the lowest, about 0.15%. In possessor condition, *ziji* is the most frequent pronoun among three alternatives in *BA*-construction, about 71.29%. The frequency of *ta* in coargument condition in *BA*-construction is about 28.45%. The frequency of *taziji* in coargument condition in *BA*-construction is least frequent one, about 0.26%.

Table 3
Corpus Study Result

Pronoun	Coargument	Possessor
ziji	29.54%	71.29%
ta	70.31%	28.45%
taziji	0.15%	0.26%
Sum	100%	100%

4 Results

This section presents the results of fitting Bayesian and RSA models to the experimental data. How well does a Bayesian Model capture the relationship between production and interpretation for reflexive pronouns in Mandarin Chinese? What about the RSA model? After examining the results, here is one short description of the key finding: **Both Bayesian Models and RSA Models consistently do a good job (p-value is less than 0.05 and Pearson correlation coefficient r-value is greater than 0.50 in a correlation analysis) when the utterance is the special ambiguous pronoun *ziji*, while both models only perform well in some of *ta* conditions, the other ambiguous pronoun.**

In the rest of this section, I will present the results in more detail. For each kind of models, I will first revisit the core ideas. Then I will introduce how to build the models and the analysis ideas. Finally, I will show the results of the analysis.

4.1 Bayesian Models

To recap, the Bayesian Model hypothesizes that pronoun interpretation is driven by the combination of two sets of biases, listener's prior world knowledge bias and speaker's production bias, as captured by the Bayesian rule in (4), repeated below as (27):

(27) $P_L(referent \mid utterance) \propto P(referent) \times P_S(utterance \mid referent)$

The prior $P(referent)$ is listeners' world knowledge bias on which referent is more likely to be mentioned in the specific communication context, regardless of the referring expression used. This was quantified in the World Knowledge Bias Test experiment. The likelihood, $P_S(utterance \mid referent)$, represents the speakers' problem on deciding which pronoun to use to refer to a certain referent. This was quantified in the Pronoun Production Task experiment. In the Bayesian model, a listener's interpretation bias relies jointly on the prior knowledge that a particular referent will be more likely to be mentioned than other alternatives under a specific context and on the likelihood that the speaker would have chosen a specific pronoun to convey the information of that referent. So knowing the prior and likelihood, the Bayesian Models are able to generate predictions on the posterior, $P_L(referent \mid utterance)$, which represents the listeners' problem on guessing what the intended referent is for a given pronoun. If Bayesian Models are correct, there should be a strong correlation between the model predictions and the experimental data which is obtained from the Pronoun Interpretation Task experiment.

Therefore, I apply Bayesian Model to predict the interpretation of each item by taking the product of the item's prior and likelihood[2]. And then I establish a correlation analysis between the model prediction and experimental data. The results are shown in Figure 11, Figure 12, and Figure 13. For all scatter plots, the horizontal coordinate represents data from the Pronoun Interpretation Task experiment and the vertical

[2] The code is available in my GitHub:
`https://github.com/zhaofylisa/Pronoun-Resolution-in-Mandarin`.

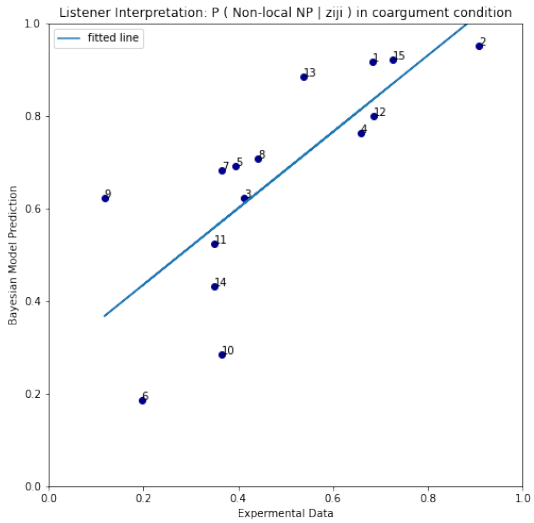
coordinate shows Bayesian model prediction of the probability of an interpretation given a specific pronoun. Each data point (each dot) in scatter plots represents a stimuli. The straight line through each scatter plot refers to line of best fit that best expresses the relation between data point. The best fit line are produced by the least squares method to arrive at the geometric equation for the line.

ziji. Figure 11 is pronoun interpretation results for Bayesian Model when the pronoun is the special reflexive *ziji*. Since *ziji* has two possible referents — Non-local NP and Local NP — and there are two types of stimuli — coargument and possessor, there are four conditions: $P(Non-localNP|ziji)$ in coargument condition, $P(Non-localNP|ziji)$ in possessor condition, $P(LocalNP|ziji)$ in coargument condition, and $P(LocalNP|ziji)$ in possessor condition. For all conditions, there are strong correlations between experimental data and Bayesian model prediction after calculating Pearson correlation coefficient r value and conducting t-test ($r > 0.77$, $p < 0.001$).

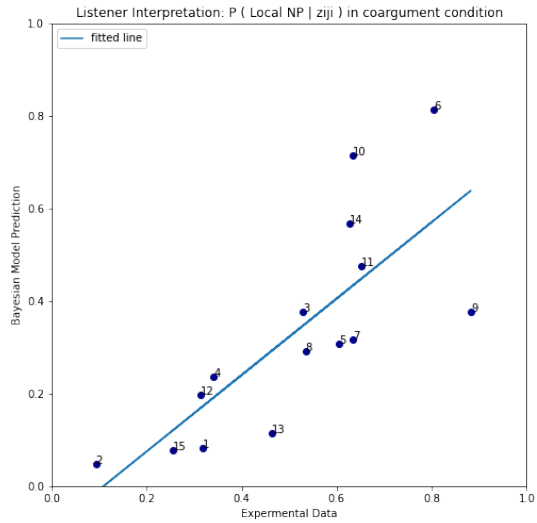
Take the first result plot Figure 11(i) as an example ($P(Non-localNP|ziji)$ in coargument condition). The mean and standard deviation of experimental data are 0.479 and 0.208 respectively. The mean and standard deviation of Bayesian model prediction are 0.667 and 0.222 respectively. Although the probabilities predicted by the model were generally higher than the actual experimental data, we still can say that Bayesian Models perform well because it has a very strong positive correlation with experimental data ($r = 0.772$, $p = 0.0007$). $P(Non-localNP|ziji)$ in Figure 11(iii) in possessor condition has a similar result ($r = 0.778$, $p = 0.0006$). Since $P(LocalNP|ziji)$ is complementary to $P(Non-localNP|ziji)$, in Figure 11(ii) and Figure 11(iv) the Bayesian models underpredict the interpretation probabilities. But there still exists strong positive correlations between experimental data and model predictions ($r = 0.770$, $p = 0.0007$; $r = 0.774$, $p = 0.0007$).

Figure 11

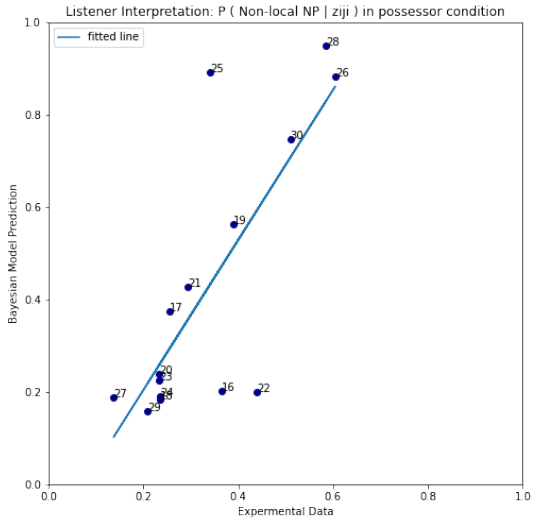
Results for Bayesian Model (pronoun = ziji)



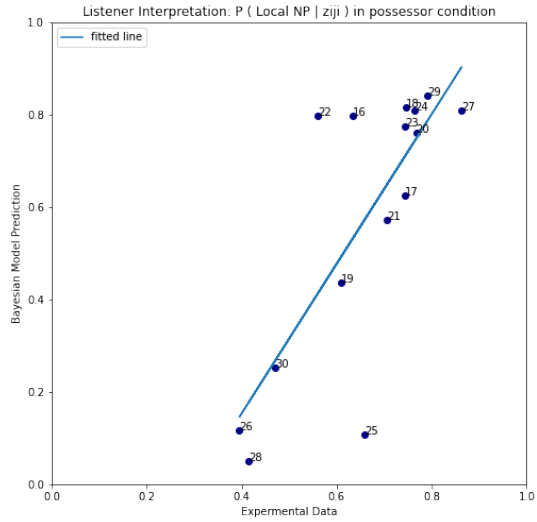
(i) *P(Non-local NP | ziji) in coargument condition*



(ii) *P(Local NP | ziji) in coargument condition*



(iii) *P(Non-local NP | ziji) in possessor condition*



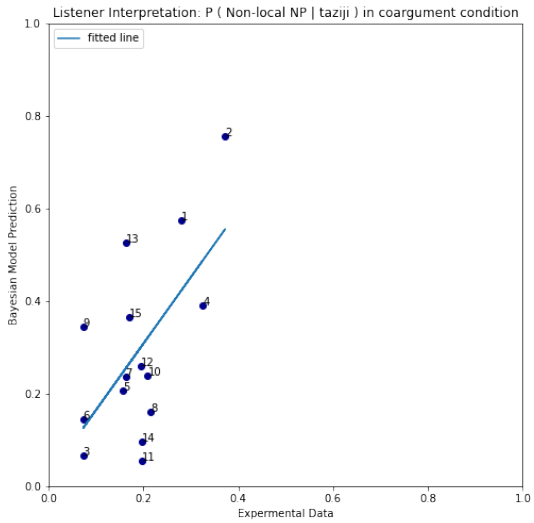
(iv) *P(Local NP | ziji) in possessor condition*

taziji. Figure 12 is pronoun interpretation results for Bayesian Model when the pronoun is *taziji*. Although this pronoun is an unambiguous one, there were still some ungrammatical selections in the experiments. This gives us some data points that can test whether Bayesian Models can explain these ungrammatical interpretations. According to the experiment result, in addition to the grammatical referent — Local NP, participants sometimes choose Non-Local NP as the antecedent of *taziji*. Combining with two types of stimuli — coargument and possessor, there are four conditions as well: $P(Non-localNP|taziji)$ in coargument condition, $P(Non-localNP|taziji)$ in possessor condition, $P(LocalNP|taziji)$ in coargument condition, and $P(LocalNP|taziji)$ in possessor condition. For all conditions, there are strong correlations between experimental data and Bayesian model prediction after calculating Pearson correlation coefficient r value and conducting t-test ($r > 0.63$, $p < 0.012$).

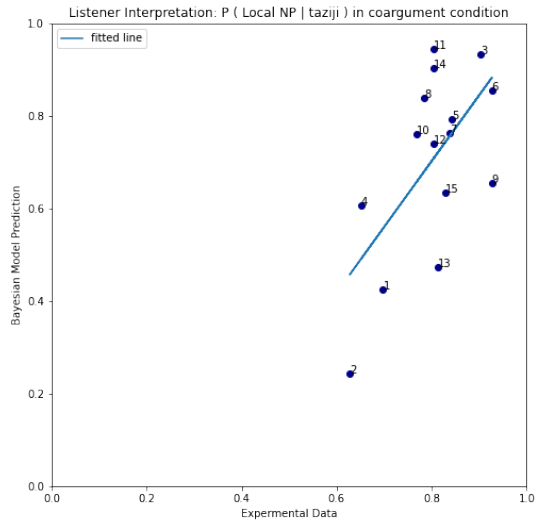
Take the first result plot Figure 12(i) as an example ($P(Non-localNP|taziji)$ in coargument condition), the mean and standard deviation of experimental data are 0.191 and 0.084 respectively, while the mean and standard deviation of Bayesian model prediction are 0.295 and 0.194 respectively. Similar to results for *ziji*, the model overpredicts the interpretation probabilities a little bit. However, we still can see that Bayesian Models perform well which has a very strong positive correlation with experimental data ($r = 0.617$, $p = 0.014$). $P(Non-localNP|taziji)$ in Figure 12(iii) in possessor condition has a similar result ($r = 0.788$, $p = 0.0004$). Since $P(LocalNP|taziji)$ is complementary to $P(Non-localNP|taziji)$, in Figure 12(ii) and Figure 12(iv) the Bayesian models underpredict the interpretation probabilities a little bit. But there still exists strong positive correlations between experimental data and model predictions ($r = 0.629$, $p = 0.012$; $r = 0.756$, $p = 0.001$).

Figure 12

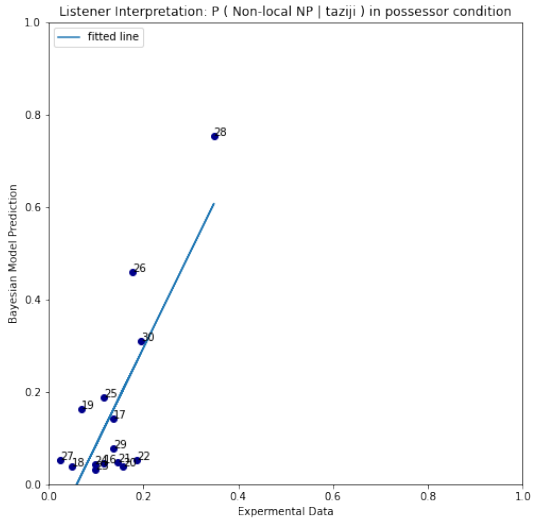
Results for Bayesian Model (pronoun = taziji)



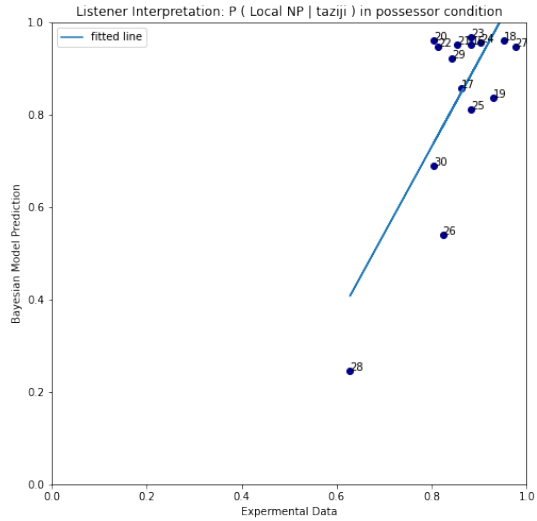
(i) *P(Non-local NP | taziji) in coargument condition*



(ii) *P(Local NP | taziji) in coargument condition*



(iii) *P(Non-local NP | taziji) in possessor condition*



(iv) *P(Local NP | taziji) in possessor condition*

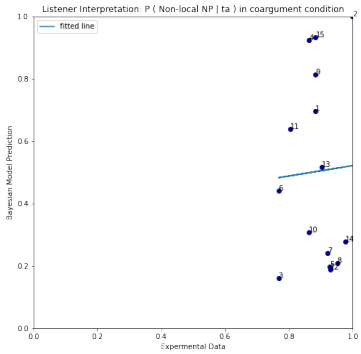
ta. Figure 13 is pronoun interpretation results for Bayesian Model when the pronoun is *ta*. Pronoun *ta* can refer to the Non-local NP, the Local NP, or Others in coargument condition. In possessor condition, *ta* can refer to the Non-local NP or the Local NP. Our experimental results show that there are some ungrammatical selection in possessor condition where participants select the Local NP as the referent to *ta*. Therefore, there are six conditions for pronoun *ta*: $P(Non-localNP|ta)$ in coargument condition, $P(Non-localNP|ta)$ in possessor condition, $P(LocalNP|ta)$ in coargument condition, $P(LocalNP|ta)$ in possessor condition, $P(Others|ta)$ in coargument condition, and $P(Others|ta)$ in possessor condition. After calculating Pearson correlation coefficient r value and conducting t-test, I find that the Bayesian model only performs well in certain conditions (in four out of six conditions): $P(Non-local|ta)$, $P(LocalNP|ta)$, and $P(Others|ta)$ in possessor condition, and $P(LocalNP|ta)$ in coargument condition. In general, the correlation between the experimental data and model predictions for the pronoun *ta* was not as strong as for the other pronouns.

There are two out of six conditions that Bayesian Models do not fit the experimental data. The first one is $P(Non-localNP|ta)$ in coargument condition, shown in Figure 13(i). The experimental data are all concentrated at 0.8 while the model predictions have a large deviation between 0.2 and 0.9. There is no correlation between the experimental data and model predictions with a statistical significance ($r = 0.037$, $p = 0.895$). The other condition when the Bayesian Models do not perform well is $P(Others|ta)$ in coargument condition. The mean and standard deviation of experimental data are 0.033 and 0.022 respectively, while the mean and standard deviation of Bayesian model prediction are 0.650 and 0.170 respectively. The models make great overpredictions and there is not correlation between experimental data and model predictions ($r = 0.328$, $p = 0.354$).

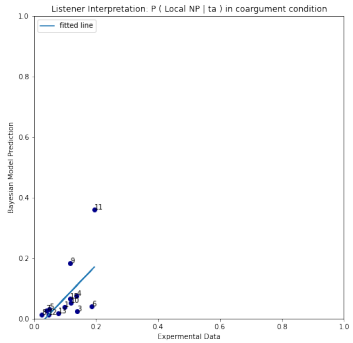
For other four conditions, the Bayesian Models perform relatively well. Although $P(LocalNP|ta)$ in coargument condition is an ungrammatical pair, there is a positive correlation between experimental data and model prediction ($r = 0.598$, $p = 0.031$). The

correlations are also found in three other possessor conditions: $P(Non-local|ta)$ ($r = 0.571$, $p = 0.026$), $P(LocalNP|ta)$ ($r = 0.658$, $p = 0.011$), and $P(Others|ta)$ ($r = 0.533$, $p = 0.023$).

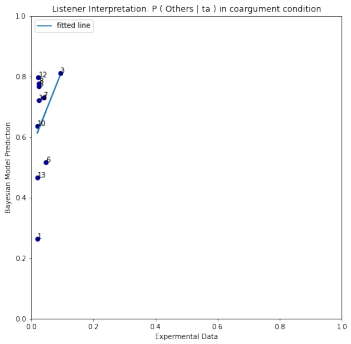
Figure 13
Results for Bayesian Model (pronoun = ta)



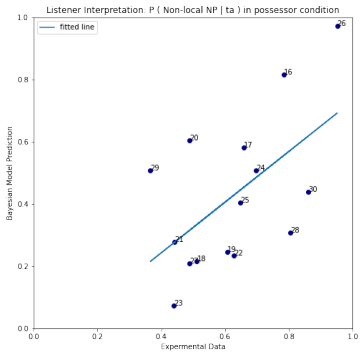
(i) *P(Non-local NP | ta) in coargument condition*



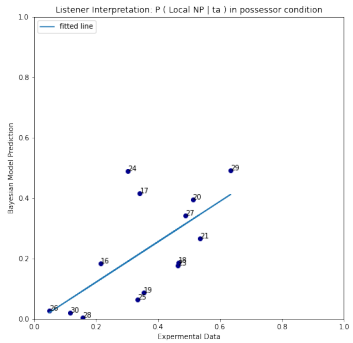
(ii) *P(Local NP | ta) in coargument condition*



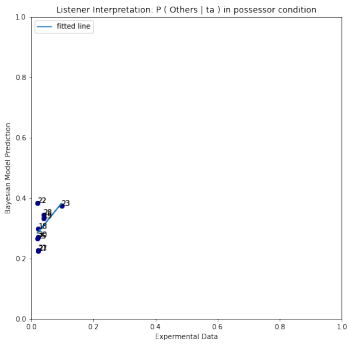
(iii) *P(Others | ta) in coargument condition*



(iv) *P(Non-local NP | ta) in possessor condition*



(v) *P(Local NP | ta) in possessor condition*



(vi) *P(Others | ta) in possessor condition*

4.2 RSA Models

To recap, the Rational Speech Act Model hypothesizes a rational listener assumes the speaker has already chosen the best utterance that a literal listener would correctly infer the referent given the literal meaning of the utterance, and combines this assumption with their prior world knowledge to interpret an utterance, as captured by a system of equations in (10), repeated below as (28):

(28)
1. Rational Listener:

   $P_L(referent \mid utterance) \propto P(referent) \times P_S(utterance \mid referent)$
2. Speaker Decision-making (Softmax Choice):

   $P_S(utterance \mid referent) \propto exp(\alpha \times U_S(utterance,\ referent))$
3. Speaker Utility:

   $U_S(utterance,\ referent) = log(P(referent \mid utterance) + C(utterance)$
4. Literal Listener:

   For all $utterance \in pronouns$ and $referent \in referents$,

   $P_{L_0}(referent \mid utterance) = \frac{P(referent \cap utterance)}{P(utterance)}$
5. Cost Function:

   $C(utterance) = \log(P(utterance))$

Similar to Bayesian Models, the prior $P(referent)$ is part of the pronoun interpretation process in RSA models. Therefore, results of the World Knowledge Bias Test experiment are used in RSA model construction. Since the likelihood $P_S(utterance \mid referent)$ can be calculated if we know $P(utterance)$ and grammatical rules, results of the Pronoun Interpretation Task experiment are not needed. The frequency of an utterance — $P(utterance)$ — can be estimated from the results of the corpus study. If RSA Models are correct, there should be a strong correlation between model predictions and experimental data which is obtained from the Pronoun Interpretation Task experiment.

As a result, I apply RSA Model to predict the interpretation of each item by calculating the system of equations in (28) [3]. And then I establish a correlation analysis between model predictions and experimental data. The results are shown in Figure 14 and Figure 15. The construction of scatter plots is similar to Bayesian Models: the horizontal coordinate represents data from the Pronoun Interpretation Task experiment and the vertical coordinate shows RSA model prediction of the probability of an interpretation given a specific pronoun. Each data point represents a stimuli. The straight line through each scatter plot refers to line of best fit produced by the least squares method. Two things to note are that there are no results for the interpretation of *taziji* and there are no results for ungrammatical pronoun interpretation for RSA Models. Since RSA Model integrates grammatical binding rules in the Literal Listener process in Equation (28-4), it only has quantitative analysis for grammatical and ambiguous pronoun interpretation process. In other words, it predicts a zero probability for ungrammatical pronoun interpretation such as $P_S(Non-localNP|taziji)$, and predicts a probability of one for grammatical and unambiguous pronoun interpretation such as $P_S(LocalNP|taziji)$.

ziji. Figure 14 is pronoun interpretation results for RSA Model when the pronoun is the special reflexive *ziji*. Since *ziji* has two possible referents — Non-local NP and Local NP — and there are two types of stimuli — coargument and possessor, there are four conditions: $P(Non-localNP|ziji)$ in coargument condition, $P(Non-localNP|ziji)$ in possessor condition, $P(LocalNP|ziji)$ in coargument condition, and $P(LocalNP|ziji)$ in possessor condition. For all conditions, there are strong correlations between experimental data and Bayesian model prediction after calculating Pearson correlation coefficient r value and conducting t-test ($r > 0.77$, $p < 0.001$).

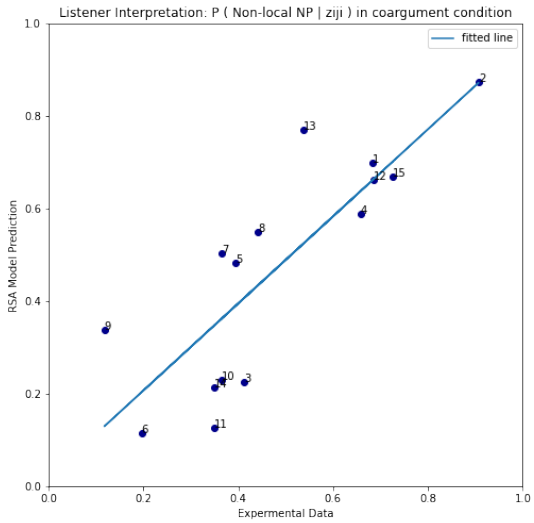
The first result plot, Figure 14(i), is a result for interpreting the pronoun *ziji* as the Non-local NP in coargument condition. The mean and standard deviation of experimental

[3] The code is available in my GitHub: `https://github.com/zhaofylisa/Pronoun-Resolution-in-Mandarin`.

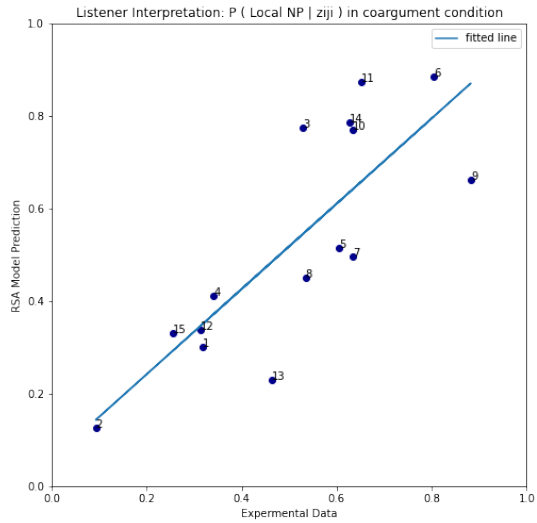
data are 0.479 and 0.208 respectively, while the mean and standard deviation of RSA model prediction are similar with 0.470 and 0.238 respectively. The RSA Models perform well because its predictions have a very strong positive correlation with experimental data ($r = 0.821$, $p = 0.0001$). $P(Non-localNP|ziji)$ in possessor condition in Figure 14(iii) represents the probability of taking the Non-local NP as the referent of *ziji* in possessor condition. The model (mean = 0.408, sd = 0.277) overpredicts a little bit higher than experimental data (mean = 0.338, sd = 0.138), but the correlation is strong ($r = 0.803$, $p = 0.0003$). Figure 14(ii) represents a result for interpreting the pronoun *ziji* as the Local NP in coargument condition. Model predictions (mean = 0.530, sd = 0.238) have similar distribution with experimental data (mean = 0.512, sd = 0.207). And they also have strong positive correlation ($r = 0.847$, $p = 0.00006$). Figure 14(iv) is a result for interpreting the pronoun *ziji* as the Local NP in possessor condition. The mean and standard deviation of experimental data are 0.658 and 0.137 respectively, while the mean and standard deviation of RSA model prediction are similar with 0.592 and 0.277 respectively. The correlation between the experimental data and RSA model predictions is strong ($r = 0.843$, $p = 0.00008$).

Figure 14

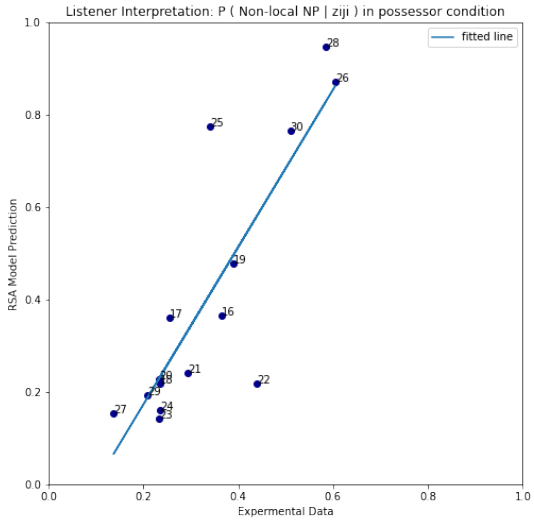
Results for RSA Model (pronoun = ziji)



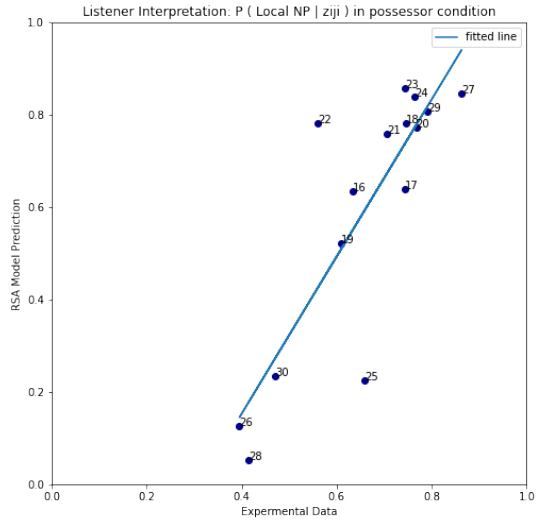
(i) *P(Non-local NP | ziji) in coargument condition*



(ii) *P(Local NP | ziji) in coargument condition*



(iii) *P(Non-local NP | ziji) in possessor condition*



(iv) *P(Local NP | ziji) in possessor condition*

ta. When the pronoun is the ambiguous *ta*, can RSA Models account for people' s interpretation preference? Results are shown in Figure 15. According to Chinese grammar rules, *ta* has three possible referents in possessor condition — Non-local NP, Local NP, and Others. It has two possible referents in coargument condition — Non-local NP and Others. Therefore, there are five conditions in total for pronoun *ta*: $P(Non-localNP|ta)$ in possessor condition, $P(LocalNP|ta)$ in possessor condition, $P(Others|ta)$ in possessor condition, $P(Non-localNP|ta)$ in coargument condition, , and $P(Others|ta)$ in coargument condition. RSA Models perform well only in two out of five conditions.

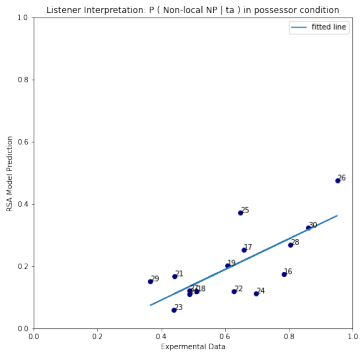
RSA Models perform well in Figure 15(i) and Figure 15(ii). Figure 15(i) represents the result for interpreting *ta* as the Non-local NP in possessor condition. The model (mean = 0.202, sd = 0.112) underpredicts the resolution process than experimental data (mean = 0.625, sd = 0.166). However, the correlation between experimental data and model predictions is strong ($r = 0.731$, $p = 0.002$). Figure 15(ii) represents the result for interpreting *ta* as the Local NP in possessor condition. Model prediction (mean = 0.351, sd = 0.203) and experimental data (mean = 0.355, sd = 0.161) have similar distribution. The result reveals a strong correlation between experimental data and model predictions ($r = 0.767$, $p = 0.0008$).

RSA Models perform poorly in Figure 15(iii), Figure 15(iv), and Figure 15(v). Figure 15(iii) shows probability of taking other person who is not in the context as the referent of *ta* in possessor condition. The probability (mean = 0.021, sd = 0.025) in experimental data is much lower than model predictions (mean=0.447, sd=0.144). The correlation between experimental data and model predictions is not significant ($r = 0.455$, $p = 0.088$). In Figure 15(iv), the model underpredicts the probability of interpreting *ta* as the Non-local NP in coargument condition (experiment: mean = 0.888, sd = 0.066; model: mean = 0.373, sd = 0.175). The correlation is not significant, and the r value is even negative ($r = -0.180$, $p = 0.521$). Figure 15(v) shows that the model overpredicts the probability of interpreting *ta* as other person who is not in the context in coargument

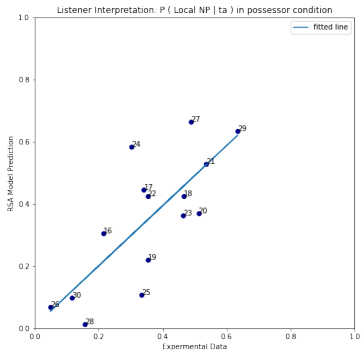
condition (experiment: mean = 0.022, sd = 0.024; model: mean = 0.627, sd = 0.175). The correlation between experimental data and model prediction is not strong ($r = 0.325$, $p = 0.237$).

Figure 15

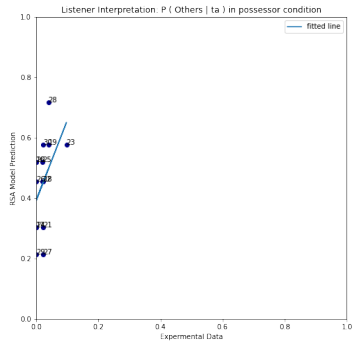
Results for RSA Model (pronoun = ta)



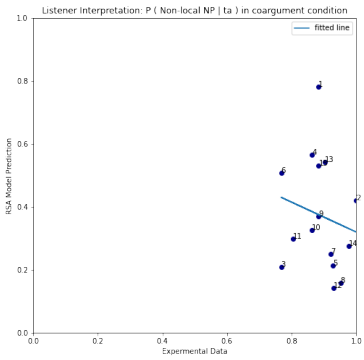
(i) *P(Non-local NP | ta) in possessor condition*



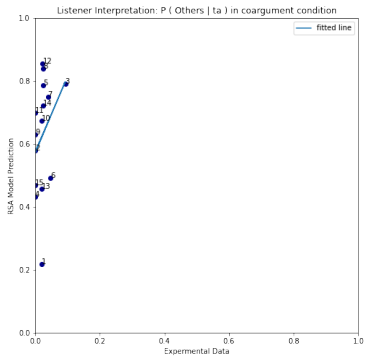
(ii) *P(Local NP | ta) in possessor condition*



(iii) *P(Others | ta) in possessor condition*



(iv) *P(Non-local NP | ta) in coargument condition*



(v) *P(Others | ta) in coargument condition*

5 Discussion

This study presented new corpus and experimental studies of pronoun interpretation in Mandarin Chinese. The empirical data showed us that the pronoun interpretation process is not only directional and qualitative, but also graded and quantitative. In other words, the pronoun interpretation is not just about choosing one character as the referent of a pronoun, but also about the strength of the preference when choosing the referent: when the preference of referent is strong and when is weak? Different context and languages may cause different interpretation preferences or different strength of one preference. For example, the interpretation of pronoun in sentence "the postman met the streetsweeper before he went home" differs across languages. Another example is that the verb "betrayed" causes a stronger preference to Non-local NP in the interpretation process of pronoun *ziji* than the verb "confused" according to our empirical results.

The Bayesian model and RSA model as probabilistic approaches to pronoun interpretation are two attempts to capture this quantitative characteristics of our experimental data. They represent two kinds of hypotheses about pronoun interpretation process. We first tested the hypotheses that pronoun interpretation preferences can be derived from The combined effect of listeners' prior world knowledge and speakers' pronoun production biases. This test is with the help of building and applying Bayesian Model. Then, we tested the hypotheses that a rational listener interprets an utterance by assuming the speaker has already chosen the best utterance among all alternative pronouns (based on the utility of an utterance), combining with their prior world knowledge. This recursive thinking between listeners and speakers was built in RSA Models in an explicit manner. In both models, the inferences of the listeners can be formed not only on the utterance that was generate by the speakers, but also on possible alternatives that were not used by the speakers.

The correlation analysis in the previous section showed that the Bayesian and RSA models can fit most of the experimental data well, except in two conditions when pronoun

Table 4
Bayesian Model Results

	P(Non-local NP|ziji)		P(Local NP|ziji)		P(Non-local NP|taziji)		P(Local NP|taziji)	
	CO [a]	PO[b]	CO	PO	CO	PO	CO	PO
r	0.772	0.778	0.770	0.774	0.617	0.788	0.629	0.756
p	**	**	**	**	*	**	*	**
	P(Non-local NP|ta)		P(Local NP|ta)		P(Others|ta)			
	CO [a]	PO[b]	CO	PO	CO	PO		
r	0.037	0.571	0.598	0.658	0.328	0.533		
p	0.895	*	*	*	0.354	*		

Note. [a]CO: coargument condition [b]PO: possessor condition
$^{*}p < .05$. $^{**}p < .01$.

is *ta* (P(Others | ta) in coargument condition and P(Non-local NP | ta) in coargument condition), both models fail. The results are organized in Table 4 and Table 5. While above two hypothesises that tested by two models are correct in most of the conditions, they make different claims about the interpretation process. By taking a look into the results for two models, we may infer which one is more correct than the other in certain conditions. RSA model performs better than Bayesian model when interpreting *ziji* with lower p-value and higher correlation coefficient. RSA model also performs better than Bayesian model when interpreting *ta* as the Non-local NP and as the Local NP in possessor condition. In addition, the Bayesian model generates predictions by an amount that exceeds (overprediction) or is too small for (underprediction) the actual experimental data. The RSA model does not severely have these overprediction or underprediction issues. The Bayesian Model performs better than RSA Model only in one condition: interpreting *ta* as Others in possessor condition. The RSA Model does not yield a statistically significant correlation with the data, but the RSA model does show a modest correlation (r = 0.533, p

Table 5
RSA Model Results

	P(Non-local NP|ziji)		P(Local NP|ziji)		P(Non-local NP|ta)		P(Local NP|ta)	P(Others|ta)	
	CO [a]	PO[b]	CO	PO	CO	PO	PO	CO	PO
r	0.821	0.847	0.803	0.843	-0.18	0.731	0.767	0.325	0.455
p	**	**	**	**	0.521	**	**	0.237	0.088

Note. [a]CO: coargument condition [b]PO: possessor condition
$^{*}p < .05$. $^{**}p < .01$.

= 0.023). Technically speaking, the RSA model makes a more restrictive hypothesis than Bayesian Model since RSA Model is built on Bayesian Model by adding more constraints to the likelihood term. Nevertheless, RSA Model still performs well and even better than the Bayesian Model. This points us in a direction for understanding listeners' reasoning when resolving ambiguous pronouns — listeners are actually probabilistic and this process is well modeled by a recursive reasoning process between speakers and listeners.

Limitations and Future Work. We cannot ignore the poor performance of the model when the pronoun is *ta*. There is a limitation in design that may cause this poor performance. *ta* as a personal pronoun is able to refer to either a person in the stimuli or a person that is not in the given stimuli (i.e., a person in the broader context). Therefore, the interpretation of *ta* may have a high demand on the broader context when it refers to other person, while other reflexive pronouns do not have this demand.

This special demand of *ta* cannot be provided in the experiment due to the limitation of the length of stimuli. This may lead to a possibility that the Pronoun Interpretation Task does not fully capture the true posterior for *ta*: Although in the real life it is common to use *ta* to refer to other people that is in the broader context, participants in my experiment only interpreted *ta* as characters that in the stimuli and seldom referred it to other people due to the lack of context. Therefore, for future study, the task to test the true interpretation of *ta* can be improved.

Another limitation is that the RSA Model is not able to fit ungrammatical data so far since it strictly incorporates grammar rules in Mandarin Chinese. A looser version of the RSA Model could be built to allow ungrammatical interpretation since they do exist in our empirical findings. A possible way to do so is to assign different weights to different pairs. An ungrammatical pair will have a small weight so it is still possible for an interpretation but is not equally likely as other grammatical pairs.

Moreover, although this study tests two probabilistic models to capture the variation across items, I did not explain why there exists this variation. It would be interesting to investigate on the action verbs used in the stimuli. Why do some of the action verbs have a very strong Non-local NP preference? Why do some of them have a weak Local NP preference? Are there any underlying shared properties of verbs that share similar preference directions? And how are these underlying properties represented in both Bayesian Model and RSA Model? I will leave these questions for future investigation.

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Appendix A

Coargument Stimuli[4]

1. 把自己弄糊涂了。confused *ziji*
2. 把自己出卖了。betrayed *ziji*
3. 把自己照顾得很好。took good care of *ziji*
4. 把自己当成小孩子了。treated *ziji* as a child
5. 把自己弄伤了。hurted *ziji*
6. 把自己想像成侦探了。imagined *ziji* as a policeman
7. 把自己灌醉了。got *ziji* drunk
8. 把自己绊倒了。tripped over *ziji*
9. 把自己锁在教室里了。locked *ziji* in the classroom
10. 把自己逼得太紧了。pushed *ziji* too hard
11. 把自己置于危险中。put *ziji* in danger
12. 把自己气哭了。made *ziji* cry
13. 把自己吓坏了。scared *ziji*
14. 把自己弄骨折了。broken *ziji*'s bones
15. 把自己逗笑了。made *ziji* laugh

[4] The pronoun *ziji* was used as example in the Stimuli.

Appendix B

Possessor Stimuli

1. 把自己的照片发到了朋友圈。posted *ziji*'s photos to social media
2. 把自己的演讲稿修改了。revised *ziji*'s speech
3. 把自己的想法忘记。forgot *ziji*'s ideas
4. 把自己的作业遗漏了。left out *ziji*'s homework
5. 把自己的遭遇讲给了别人听。told others *ziji*' s encounters
6. 把自己的工作做完了。finished *ziji*'s work
7. 把自己的想法表达出来了。expressed *ziji*'s idea
8. 把自己的孩子搂在了怀里。held *ziji*'s child in arms
9. 把自己的证件弄丢了。lost *ziji*'s documents
10. 把自己的小狗找回来了。found *ziji*'s dog
11. 把自己的玩具模型弄坏了。broke *ziji*'s toy model
12. 把自己的房间打扫干净了。cleaned *ziji*'s room
13. 把自己的秘密告诉了别人。told *ziji*'s secret to others
14. 把自己的衣服捐掉了。donated all *ziji*'s clothes
15. 把自己的文件删除了。deleted *ziji*'s documents